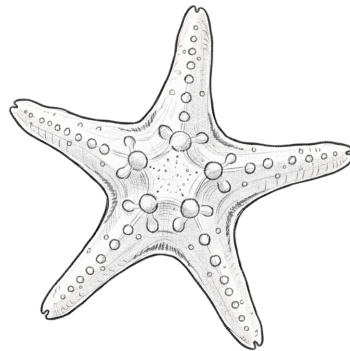


Elementary Deep Learning

Research Training Project VI

Yimin Zhong



*Department of Mathematics and Statistics,
Auburn University*

Preface

This short note is the sixth component of the *Research Training Project* (RTP). The purpose is to provide an introduction to deep learning from a mathematical perspective. The note is drafted as *self-contained* as possible with explanatory examples, which may sacrifice some rigor and depth.

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Part I

Preliminaries

Chapter 0

Expedition for Alchemists

It's the possibility of having a dream come true that makes life interesting.

– Paulo Coelho, [The Alchemist](#)



EEP learning has been one of the most popular topics in science and engineering over the past decade and is undergoing rapid advances. This chapter aims to provide beginners, especially “alchemists” who are less concerned with mathematical principles, with a superficial yet mathematically rigorous overview of the essential elements of deep learning.

0.1 Brief Introduction

In brief, *deep learning* is a paradigm based on composed functions of simple functions, called *neural networks*. The composition of multiple (layers) simple functions provides sufficient capacity for representation and flexibility for nonlinear approximation in high dimensions.

The *neural networks* were previously studied by computer scientists, but the real impact of deep learning was revealed only in the last decade. With the advent of GPU architectures, deep learning approaches have begun to outperform classical algorithms in computer vision, data mining, natural language processing, robotics, and related areas. However, this note does not emphasize the practical elements of computation. Instead, we will focus more on the mathematical principles, using basic tools from linear algebra, differential equations, and probability.

0.2 Preparations

For simplicity, we view neural networks as maps from the data space $\mathcal{X}$ to the label space $\mathcal{Y}$. The dependent parameters are denoted by $\theta \in \Theta$, where Θ is an admissible set embedded in some other space. The abstract structure of the neural networks is $f : \mathcal{X} \times \Theta \rightarrow \mathcal{Y}$, and the objective is to find a parameter $\theta^* \in \Theta$ such that $f(x, \theta^*)$ achieves a specific goal.

In the context of supervised learning, the ground-truth datapoints $(x_k, y_k) \in \mathcal{D} \subset \mathcal{X} \times \mathcal{Y}$ are provided, and the objective is to approximate a function (neural network) $f(x, \theta^*)$ that can be used to predict the labels for unseen samples in $\mathcal{X}$. The parameter θ^* is found by minimizing the (average) loss function.

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} \ell(y, f(x, \theta)),$$

where $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+$ such that $\ell(y, y') = 0$ if and only if $y = y'$, but ℓ is not necessarily a metric or quasi-metric. Typically, the choice of the loss function ℓ and the model f can significantly affect performance and may induce biases.

Remark 0.2.1. In unsupervised learning, labels are unavailable (partly available in weak supervised learning), and only the input data $x_k \in \mathcal{X}$ are given. The objective is to learn the hidden patterns from the data. It is commonly used for clustering tasks, such as recommendation systems. However, it is not the focus of this note.

0.2.1 Basic Probability

We will use some basic probability theory later.

Definition 0.2.2. A probability space $(\Omega, \mathcal{F}, \mathbb{P})$ means

1. Ω is the sample space.
2. $\mathcal{F}$ is a σ -algebra or event space (set of events).
3. $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a probability measure.

Definition 0.2.3. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and E be a measurable space. A random variable $X : \Omega \rightarrow E$ is a measurable function.

$$\mathbb{P}[X \in S \subset E] = \mathbb{P}[\{\omega \in \Omega \mid X(\omega) \in S\}]$$

Definition 0.2.4. The expectation of $X : \Omega \rightarrow E$ is

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) d\mathbb{P}(\omega).$$

If $f : E \rightarrow \mathbb{R}$ is a measurable function, then

$$\mathbb{E}[f(X)] = \int_{\Omega} f(X(\omega)) d\mathbb{P}(\omega).$$

Remark 0.2.5. Suppose $\mathbb{E}[f(X)] = m$. It means there exists $\omega \in \Omega$ such that $f(\omega) \geq m$ (or $\leq m$). If the variance $\text{Var}(f(X)) \neq 0$, then the inequality is strict.

Theorem 0.2.6 (Jensen's Inequality). Suppose h is a convex function, then

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

Definition 0.2.7. The **probability density function** of a random variable X with

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respect to a reference measure μ is the Radon-Nikodym derivative

$$dX_*\mathbb{P} = f d\mu$$

and

$$\mathbb{P}[X \in S] = \int_S f d\mu$$

Definition 0.2.8. Let $p \geq 1$. The random variable $X : \Omega \rightarrow E$ satisfies $X \in L^p(\mathbb{P})$ if

$$\|X\|_p = (\mathbb{E}|X|^p)^{1/p} < \infty.$$

Note that if $p < 1$, this does not define a norm.

Theorem 0.2.9 (Interpolation). Let $p_1, p_2 \geq 1$. If the random variable $X \in L^{p_1}(\mathbb{P}) \cap L^{p_2}(\mathbb{P})$, then for any $p \in (p_1, p_2)$,

$$\|X\|_p \leq \|X\|_{p_1}^\alpha \|X\|_{p_2}^{1-\alpha},$$

where $\frac{1}{p} = \frac{\alpha}{p_1} + \frac{1-\alpha}{p_2}$.

Definition 0.2.10. Let $X \in L^2(\mathbb{P})$, the variance of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}|X|^2 - \mathbb{E}[X]^2.$$

If X is multi-dimensional, then it defines the covariance matrix

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^T].$$

Theorem 0.2.11 (Chebyshev). Let $X \in L^2(\mathbb{P})$ be a random variable, then for any $\delta > 0$,

$$\mathbb{P}(|X| > \delta) \leq \frac{\mathbb{E}|X|^2}{\delta^2}.$$

Definition 0.2.12. The conditional probability of an event $B \subset \mathcal{F}$ given the event $A \subset \mathcal{F}$ is

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)}.$$

Theorem 0.2.13 (Bayes). Let $A, B \subset \mathcal{F}$, then

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Definition 0.2.14. Let X_n be a sequence of random variables.

1. Convergence in probability: $X_n \xrightarrow{p} X$ if for all $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0.$$

2. Convergence almost surely: $X_n \xrightarrow{a.s.} X$ if

$$\mathbb{P}(\lim_{n \rightarrow \infty} X_n = X) = 1.$$

3. Convergence in L^p : $X_n \xrightarrow{L^p} X$ if

$$\lim_{n \rightarrow \infty} \mathbb{E}|X_n - X|^p = 0.$$

4. Convergence in distribution (weak convergence): $X_n \xrightarrow{d} X$ if for any continuous and bounded f ,

$$\lim_{n \rightarrow \infty} \mathbb{E}f(X_n) = \mathbb{E}f(X).$$

Theorem 0.2.15. The relationship between the convergences is well known.

$$X_n \xrightarrow{a.s.} X \Rightarrow X_n \xrightarrow{p} X \Rightarrow X_n \xrightarrow{d} X,$$

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and

$$X_n \xrightarrow{L^p} X \Rightarrow X_n \xrightarrow{p} X \Rightarrow X_n \xrightarrow{d} X.$$

Definition 0.2.16 (Tightness). Let $\mathbb{P}_n$ be a sequence of probability measures on Ω . If for any $\varepsilon > 0$ there exists n_0 and a compact set $K \subset \Omega$ that

$$\mathbb{P}_n(K) > 1 - \varepsilon, \quad \forall n > n_0.$$

Theorem 0.2.17 (Prohorov). A sequence of probability measures is tight, then it contains a weakly convergent subsequence.

Theorem 0.2.18 (Law of Large Numbers). Let X_n be a sequence of independent and identically distributed (i.i.d.) random variables $\sim X$ that $\mathbb{E}[X] = \mu < \infty$. Define

$$Z_N = \frac{1}{N} \sum_{n=1}^N X_n.$$

Then

1. $Z_N \xrightarrow{a.s.} \mu$.
2. Suppose that $\mathbb{E}|X|^p < \infty$, then $Z_N \xrightarrow{L^p} \mu$.

Remark 0.2.19. The weak law of large numbers is directly from Chebyshev's inequality. The strong version needs the Borel-Cantelli theorem. See [Durrett \(2019\)](#) for a proof.

Question 0.2.20. Prove the weak version of the LLN.

Theorem 0.2.21 (Central Limit Theorem). Let X_n be a sequence of i.i.d. random

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variables with $\mathbb{E}X = \mu$ and $\text{Var } X = \sigma^2$. Define

$$S_N = \sum_{n=1}^N X_n.$$

Then

$$\frac{S_N - N\mu}{N^{1/2}\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

Remark 0.2.22. The usual proof uses the characteristic function and the uniqueness of the Hamburger moment problem. See [Durrett \(2019\)](#) for a proof.

Theorem 0.2.23 (Lindeberg-Feller). A generalization of the central limit theorem is the following. Let $X_{n,m}$ be a triangular sequence, $1 \leq m \leq n$, be independent random variables such that $\mathbb{E}X_{n,m} = 0$. If

1. $\sum_{m=1}^n \mathbb{E}X_{n,m}^2 \rightarrow \sigma^2 > 0$;
2. For any $\varepsilon > 0$, $\lim_{n \rightarrow \infty} \mathbb{E}[X_{n,m}^2 \mathbf{1}_{|X_{n,m}| > \varepsilon}] = 0$. This is called Lindeberg's condition.

Then $S_n = \sum_{m=1}^n X_{n,m}$ converges weakly to standard normal distribution.

Remark 0.2.24. This theorem implies that a large number of small, independent events approximately follow a normal distribution. Notice that we do not require identically distributed random variables.

Corollary 0.2.25 (Lyapunov Central Limit Theorem). Let X_n be a sequence of independent random variables with $\mathbb{E}X_n = \mu_n$, $\text{Var}(X_n) = \sigma_n^2$, and

$$\lim_{n \rightarrow \infty} \frac{1}{s_N^{2+\delta}} \sum_{n=1}^N \mathbb{E}[|X_n - \mu_n|^{2+\delta}] = 0,$$

where $s_N^2 = \sum_{n=1}^N \sigma_n^2$. Then

$$\frac{1}{s_N} \sum_{n=1}^N (X_n - \mu_n) \xrightarrow{d} \mathcal{N}(0, 1).$$

With i.i.d. random variables, the convergence rate of the central limit theorem

can be quantified.

Theorem 0.2.26 (Petrov (2012)). Let X_n be a sequence of i.i.d. random variables with $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, and $\mathbb{E}|X|^{2+\delta} = \rho < \infty$, $\delta \in (0, 1]$. Define

$$Z_N = \frac{1}{N}(X_1 + \cdots + X_N).$$

Then the cumulative distribution function of $\sqrt{N}Z_N$, denoted by F_N ,

$$F_N(x) := \mathbb{P}[x < \sqrt{N}Z_N]$$

satisfies

$$|F_N(x) - \Phi(x)| \leq C\rho N^{-\frac{\delta}{2}},$$

where Φ is the cumulative distribution function of $\mathcal{N}(0, 1)$, and C is an absolute constant.

Remark 0.2.27. The supremum of the differences between the cumulative distribution functions is known as the Kolmogorov–Smirnov test statistic.

Question 0.2.28. Use Bernoulli random variables to show that the rate in Theorem 0.2.26 does not apply for $\delta > 1$. In other words, $\delta = 1$ is sharp.

When $\delta = 1$, it becomes the Berry-Esseen theorem, which has variants for weakly independent random variables. The convergence rate of $\mathcal{O}(N^{-1/2})$ is optimal.

Theorem 0.2.29 (Berry-Esseen). Let X_n be a sequence of i.i.d. random variables with $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, and $\mathbb{E}|X|^3 = \rho < \infty$. Define

$$Z_N = \frac{1}{N}(X_1 + \cdots + X_N).$$

Then the cumulative distribution function of $\sqrt{N}Z_N$, denoted by F_N satisfies

$$|F_N(x) - \Phi(x)| \leq \frac{C\rho}{\sqrt{N}},$$

where Φ is the cumulative distribution function of $\mathcal{N}(0, 1)$, and $C < 0.4748$ is an absolute constant.

The Berry-Esseen theorem describes the cumulative distribution of the scaled variable $\sqrt{N}Z_N$, whose variance is normalized. The next theorem directly studies the convergence of the empirical cumulative distribution.

Theorem 0.2.30 (Dvoretzky–Kiefer–Wolfowitz Inequality). Let X_n be a sequence of i.i.d. random variables. The empirical cumulative distribution function F_N is

$$F_N(x) = \frac{1}{N} \sum_{n=1}^N \mathbf{1}(X_i \leq x),$$

satisfies

$$\mathbb{P}[\sup_{x \in \mathbb{R}} |F_N(x) - F(x)| > \varepsilon] \leq C e^{-2N^2\varepsilon}, \quad \forall \varepsilon > 0.$$

The constant $C = 2$ is sharp. If $\varepsilon > \sqrt{\frac{\log 2}{2N}}$, the constant $C = 1$ suffices.

Remark 0.2.31. We consider a simple example where X_n are i.i.d. random variables $\sim X$ with a uniform bound, then

$$Z_N - \mathbb{E}[X] = \int_{\mathbb{R}} x dF_N(x) - \int_{\mathbb{R}} x dF(x) = - \int_{\mathbb{R}} (F_N(x) - F(x)) dx,$$

Therefore, for any $\varepsilon > 0$, $\mathbb{P}[\lim_{N \rightarrow \infty} |Z_N - \mathbb{E}[X]| < \varepsilon] = 1$, that is, $Z_N \xrightarrow{a.s.} \mathbb{E}[X]$.

Remark 0.2.32. The multivariate version of DKW inequality can be found in [Naaman \(2021\)](#), where the constant $C = d(N + 1)$ in d dimensions. For sufficiently large N , C can be reduced to $2d$, which is sharp.

Next, we introduce two useful concentration inequalities. These bounds are sharper than Chebyshev's inequality because of the additional independence requirement.

Theorem 0.2.33 (Chernoff Bound). Let $X = \sum_{n=1}^N X_n$, where X_n are independent Bernoulli random variables with parameters p_n . Let $\mu = \mathbb{E}[X] = \sum_n p_n$. Then,

1. Upper Tail.

$$\mathbb{P}[X \geq (1 + \varepsilon)\mu] \leq \exp\left(-\frac{\varepsilon^2}{2 + \varepsilon}\mu\right), \quad \forall \varepsilon > 0.$$

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2. Lower Tail.

$$\mathbb{P}[X \leq (1 - \varepsilon)\mu] \leq \exp\left(-\frac{\varepsilon^2}{2}\mu\right), \quad \forall \varepsilon \in (0, 1).$$

A more useful version of the Chernoff bound is the following.

Remark 0.2.34. If $\varepsilon \in (0, 1)$, then

$$\mathbb{P}[|X - \mu| \leq \varepsilon\mu] \leq 2 \exp\left(-\frac{\varepsilon^2}{3}\mu\right).$$

Theorem 0.2.35 (Hoeffding's Inequality). Let X_n be independent random variables with bounds $a_n \leq X_n \leq b_n$. Let $X = \sum_n X_n$ and $\mu = \sum_n \mathbb{E}[X_n]$. Then,

$$\mathbb{P}[X \geq (1 + \varepsilon)\mu] \leq \exp\left(-\frac{2\varepsilon^2\mu^2}{\sum_n (b_n - a_n)^2}\right)$$

and

$$\mathbb{P}[X \leq (1 - \varepsilon)\mu] \leq \exp\left(-\frac{2\varepsilon^2\mu^2}{\sum_n (b_n - a_n)^2}\right).$$

Remark 0.2.36. Apply the above inequality to i.i.d. Bernoulli random variables of parameter p . It produces a two-sided tail probability

$$\mathbb{P}_{\text{Hoeffding}} = 2 \exp(-2\varepsilon^2 np^2), \quad \mathbb{P}_{\text{Chernoff}} = 2 \exp(-\varepsilon^2 np/3).$$

When p is small, the Chernoff bound is much sharper. For larger p , the bounds are comparable.

0.2.2 Basic Information Theory

The *entropy* of a random variable is the average level of *uncertainty* or *difficulty* of encoding about the variable's states. Higher entropy implies greater uncertainty and thus is more difficult to encode.

Definition 0.2.37 (Shannon Entropy). Let X be a random variable that takes values in a discrete set E . The entropy is

$$\mathbb{H}(X) = \mathbb{E}[-\log p(X)],$$

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where $p(x) = \mathbb{P}[X = x]$.

Remark 0.2.38. Note that the log function here is taken to be the natural logarithm for convenience; the output is denoted in “nats”. Usually, the log function should be $\log_2$, so the output is the real “bits”.

Remark 0.2.39. The thermodynamic entropy from statistical mechanics has a similar form (with an extra Boltzmann constant).

$$S = -k_B \sum p_i \log p_i,$$

where p_i is the probability of i -th microstate in an equilibrium ensemble. The Boltzmann entropy formula takes all p_i equal: $S = -k_B \log W$, where W is the total number of microstates.

When X is associated with a probability density function $f(x)$, then we can define its differential entropy as a generalization of the Shannon entropy.

Definition 0.2.40 (Differential Entropy). Let X be a random variable with a probability density function $f(x)$. The differential entropy is

$$\mathbb{H}(X) = \mathbb{E}[-\log f(X)] = - \int f(x) \log f(x) dx.$$

Definition 0.2.41 (Conditional Entropy).

$$\mathbb{H}(X | Y) = - \sum_{x,y} p_{X,Y}(x,y) \log \frac{p_{X,Y}(x,y)}{p_Y(y)},$$

where $p_{X,Y}$ is the joint probability, and p_Y is the marginal probability.

We can immediately find that $\mathbb{H}(X) \geq \mathbb{H}(X | Y)$ and

$$\mathbb{H}(X, Y) = \mathbb{H}(X | Y) + \mathbb{H}(Y).$$

It means that the information about X and Y jointly is equivalent to first knowing Y , then knowing X given Y (vice versa).

Definition 0.2.42. *The mutual information*

$$I(X; Y) := \mathbb{H}(X) - \mathbb{H}(X | Y) = \mathbb{H}(Y) - \mathbb{H}(Y | X).$$

Remark 0.2.43. *The mutual information is symmetric and can be written as*

$$I(X; Y) = \mathbb{H}(X) + \mathbb{H}(Y) - \mathbb{H}(X, Y)$$

Question 0.2.44. *How to use the entropy argument to estimate the upper bound of the balance puzzle?*

Remark 0.2.45. *Any deterministic processing will only reduce the entropy. Since*

$$\mathbb{H}(f(X) | X) = 0.$$

Then

$$\mathbb{H}(X) = \mathbb{H}(f(X) | X) + \mathbb{H}(X) = \mathbb{H}(f(X)) + \mathbb{H}(X | f(X)) \geq \mathbb{H}(f(X)).$$

Question 0.2.46. *A pseudorandom number generator (RNG) is a deterministic algorithm for generating random numbers from a given seed. Therefore, the entropy of the resulting sequence will not be greater than the entropy of the seed. How to understand the randomness there?*

Remark 0.2.47 (Submodularity). *Let X, Y, Z be random variables, then*

$$\mathbb{H}(X, Z) + \mathbb{H}(X, Y) \geq \mathbb{H}(X) + \mathbb{H}(X, Y, Z).$$

Since

$$\mathbb{H}(X, Z) - \mathbb{H}(X) = \mathbb{H}(Z | X) \geq \mathbb{H}(Z | X, Y) = \mathbb{H}(X, Y, Z) - \mathbb{H}(X, Y).$$

It can be readily generalized to index sets I and J .

$$\mathbb{H}(\{X_i\}_{i \in I}) + \mathbb{H}(\{X_i\}_{i \in J}) \geq \mathbb{H}(\{X_i\}_{i \in I \cap J}) + \mathbb{H}(\{X_i\}_{i \in I \cup J}).$$

Definition 0.2.48 (Relative Entropy). The Kullback-Leibler (KL) divergence of two probability distributions P and Q is defined as follows.

$$D_{KL}(P\|Q) := \sum_x P(x) \log \frac{P(x)}{Q(x)}.$$

It defines a quasi-distance (not symmetric) and measures the gain of Q over P .

Theorem 0.2.49 (Gibbs Inequality). $D_{KL}(P\|Q) \geq 0$.

Question 0.2.50. Therefore, taking Q as uniform distribution leads to $\mathbb{H}(P) \leq \log(|\Omega|)$, $|\Omega|$ is the cardinality of Ω . Prove this.

Remark 0.2.51. The mutual information $I(X; Y)$ can be written as

$$\begin{aligned} I(X; Y) &= D_{KL}(p_{X,Y} \| p_X p_Y) \\ &= \mathbb{E}_Y [D_{KL}(p_{X,Y} \| p_X)] \end{aligned}$$

The cross-entropy measures the mean extra “bits” needed for a distribution Q to encode the true distribution P .

Definition 0.2.52 (Cross-Entropy). The cross-entropy of two distributions P and Q is

$$\mathcal{C}(P, Q) = - \sum_x P(x) \log Q(x) = \mathbb{H}(P) + D_{KL}(P\|Q).$$

Remark 0.2.53. For instance, two Bernoulli distributions with parameters p and p' have the cross-entropy as

$$\mathcal{C}(p, p') = -p \log p' - (1 - p) \log(1 - p').$$

Usually, when comparing a distribution Q to a reference fixed distribution P , the cross-entropy can be regarded as the KL divergence, since $\mathbb{H}(P)$ is fixed. However, the converse (in which Q is fixed) is not equivalent.

Question 0.2.54. Let P be a uniform distribution on $[-1, 1]$ and Q be a uniform distribution on $[-2, 2]$, compute $D_{KL}(P\|Q)$ and explain why.

Empirically, the reference distribution P is not known and must be estimated from a dataset $\mathcal{D} \sim P$, which induces an empirical distribution with density $P(x) = \frac{1}{N} \sum_{x \in \mathcal{D}} \delta(x)$, where $N = |\mathcal{D}|$, that is,

$$\mathcal{C}(\mathcal{D}, Q) = -\frac{1}{N} \sum_{x \in \mathcal{D}} \log Q(x).$$

Remark 0.2.55. Finally, we note that cross-entropy is related to maximum likelihood estimation. Let $Q_\theta(x)$ be a density function where θ is a parameter to be determined, and the observation data $\mathcal{D}$ is sampled from a distribution P , that is, $\mathcal{D} \sim P$. The likelihood function is

$$\prod_{i=1}^N Q_\theta(x_i) = \exp(-N\mathcal{C}(\mathcal{D}, Q_\theta)).$$

Therefore, finding the maximum likelihood estimator $\hat{\theta}_{MLE}$ is equivalent to minimizing the cross-entropy.

0.2.3 Basic Dynamical Systems

Theorem 0.2.56 (Grönwall's Inequality). Let y be a continuous, differentiable, non-negative function on $[a, b]$ and

$$y'(t) \leq \alpha(t)y(t), \quad a < t < b.$$

Then

$$y(t) \leq y(a) \exp\left(\int_a^t \alpha(s) ds\right).$$

Remark 0.2.57. There are discrete versions and stochastic versions of Grönwall's inequality.

0.2.4 Basic Functional Analysis

Theorem 0.2.58 (Borsuk-Ulam Theorem). If $g : \mathbb{S}^d \rightarrow \mathbb{R}^d$ is a continuous odd function, then there exists an $x \in \mathbb{S}^d$ that $g(x) = \mathbf{0} \in \mathbb{R}^d$.

Theorem 0.2.59 (Baire Category Theorem). *Every completely metrizable topological space is a **Baire space**, that is, every countable intersection of open dense sets is dense in it.*

Remark 0.2.60. *The Baire Category Theorem uses the **Axiom of Choice** (AC).*

0.3 Concepts

The following concepts are useful.

0.3.1 Perceptron

A perceptron is an algorithm/function for binary classification. In a nutshell, the given data points x_k , a perceptron is described by a Heaviside (or sign) function that

$$f(x) = \text{sgn}(w \cdot x + b).$$

It identifies the hyperplane $w \cdot x + b = 0$ that separates the data points. In general, if the data points are linearly separable, the usual optimization algorithms (e.g., gradient descent) can identify a separating hyperplane rather quickly.

The multilayer perceptron (MLP) is based on the perceptron concept. MLP is also known as the *fully connected feedforward neural network*. A general MLP with $L + 1$ layers means it has L hidden layers (hidden means there are trainable variables).

$$h_j^{(l)} = \sigma(z_j^{(l)}), \quad z_j^{(l)} = \sum_{i=1}^{k_{l-1}} w_{i,j}^{(l)} h_i^{(l-1)} + b_j^{(l)}, \quad l = 1, \dots, L.$$

In the vector form, we write

$$\mathbf{h}^{(l)} = \sigma \left(\mathbf{w}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \right).$$

The weight matrix $\mathbf{w}^{(l)} \in \mathbb{R}^{k_l \times k_{l-1}}$ and $\mathbf{b}^{(l)} \in \mathbb{R}^{k_l}$, k_l is the dimension of the l -th layer. In particular, k_0 is the dimension of the input space and k_L is the dimension of the output space. The function σ is called the activation function. Common choices are ReLU (also known as the hinge function), sigmoidal, etc.

0.3.2 Regression & Classification

Regression and classification are core supervised learning tasks that handle different types of labels (continuous and discrete).

For continuous or numeric labels, the loss functions often measure the mean mismatch, e.g., MAE (absolute), MSE (squared)

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} |y - f(x, \theta)|,$$

$$\mathcal{L}_{\text{MSE}} = \frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} |y - f(x, \theta)|^2,$$

which are L^1 error or L^2 error squared with respect to the empirical probabilistic measure.

For discrete labels, the model $f(x, \theta)$ often contains a final softmax layer:

$$\text{Softmax}(z) = \frac{1}{M} (e^{z_1/T}, \dots, e^{z_n/T}), \quad M = \sum_i e^{z_i/T}.$$

to convert the numeric values into a probability distribution over the categories. T denotes the temperature relative to the Boltzmann/Gibbs distribution; the default value is $T = 1$. The loss functions are then defined as the mean/sum of the mismatch between the distributions (e.g., TV and KL divergence). In this case, the data labels are converted into one-hot vectors representing the ground-truth probabilities.

$$\mathcal{L}_{\text{TV}} = \frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} |y - f(x, \theta)|,$$

$$\mathcal{L}_{\text{KL}} = \frac{1}{|\mathcal{D}|} \sum_{(x,y) \in \mathcal{D}} y \cdot \log \frac{y}{f(x, \theta)}.$$

The KL divergence (or relative entropy) is non-symmetric and sensitive to small probabilities. Regarding the implementation, it is often directly using the cross-entropy formulation. Intuitively, the KL divergence quantifies the information gain (difference) between the modeled distribution and the ground-truth distribution, which is more physically meaningful in many tasks.

Problem 0.3.1. Why is the MSE-type loss NOT used for classification?

0.3.3 Backpropagation

The training of a neural network $f(x, \theta)$ is basically optimizing the loss function $\mathcal{L}(\theta)$ through gradient descent or its variants, which depend on the gradient information $\partial_\theta \mathcal{L}$.

Taking the MLP structure as an example. We write the neural network as a composition $f(x, \theta) = \mathbf{h}^{(L)}$ and

$$\begin{aligned}\mathbf{h}^{(l)} &= \sigma(\mathbf{w}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}), \quad l = 1, \dots, L \\ \mathbf{h}^{(0)} &= x.\end{aligned}$$

Each iteration $\mathbf{h}^{(l)}$ is produced by a single-layer perception model with weight variables $\mathbf{w}^{(l)}$ and bias variables $\mathbf{b}^{(l)}$.

For a scalar variable $\theta^{(l)}$ in the l -th layer, the corresponding gradient component is computed as

$$\begin{aligned}\frac{\partial \mathbf{h}^{(k)}}{\partial \theta^{(l)}} &= \sigma'(\mathbf{w}^{(k)} \mathbf{h}^{(k-1)} + \mathbf{b}^{(k)}) \mathbf{w}^{(k)} \frac{\partial \mathbf{h}^{(k-1)}}{\partial \theta^{(l)}}, & k > l \\ \frac{\partial \mathbf{h}^{(l)}}{\partial \theta^{(l)}} &= \sigma'(\mathbf{w}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}) \left(\frac{\partial \mathbf{w}^{(l)}}{\partial \theta^{(l)}} \mathbf{h}^{(l-1)} + \frac{\partial \mathbf{b}^{(l)}}{\partial \theta^{(l)}} \right) \\ \frac{\partial \mathbf{h}^{(k)}}{\partial \theta^{(l)}} &= 0, & k < l.\end{aligned}$$

It turns out the gradient $\frac{\partial \mathbf{h}^{(L)}}{\partial \theta^{(l)}}$ can be computed iteratively from the previous layers $k \geq l$.

One potential issue with a deep MLP model is that the gradient with respect to the innermost layers will be multiplied by σ' many times. Activation functions like logistic function $\sigma(x) = \frac{1}{1+e^{-x}}$ have a derivative strictly less than one (why?) will make the gradient $\frac{\partial \mathbf{h}^{(L)}}{\partial \theta^{(l)}}$ vanishing when $L \gg l$. Similar issues arise with ReLU activation, since a portion of the derivative is zero.

0.3.4 Regularization

Regularization is a technique for controlling training. One popular choice is to add a penalty term to the loss function.

$$\tilde{\mathcal{L}}(\theta) = \mathcal{L}(\theta) + \Omega(\theta).$$

The regularization term $\Omega(\theta)$ can be viewed as a certain prior knowledge about the parameter θ from the Bayesian viewpoint. That is, if we view the data points (x_k, y_k) are independently generated from the following model with independent

noise,

$$Y = f(X, \theta) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \eta^2).$$

Then,

$$\mathbb{P}[Y | X, \theta] \propto \exp \left(-\frac{|f(X, \theta) - Y|^2}{\eta^2} \right).$$

The maximum likelihood estimator (MLE) becomes

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \prod_{k=1}^n \exp \left(-\frac{|f(x_k, \theta) - y_k|^2}{\eta^2} \right),$$

which corresponds to the MSE loss function. We also find out

$$\mathbb{P}[\theta | Y, X] = \frac{\mathbb{P}[Y | X, \theta] \cdot \mathbb{P}[\theta]}{\mathbb{P}[Y | X]} = \frac{\text{likelihood} \cdot \text{prior}}{\text{evidence}}$$

Therefore, if we know the prior of θ , say, obeys Gaussian distribution $\mathcal{N}(0, \tau^2)$, then, the maximum *a posteriori* estimate (MAP) becomes

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \prod_{k=1}^n \exp \left(-\frac{|f(x_k, \theta) - y_k|^2}{\eta^2} \right) \cdot \exp \left(-\frac{|\theta|^2}{\tau^2} \right),$$

which corresponds to the Tikhonov regularization term $\Omega(\theta) = \frac{\eta^2}{\tau^2} |\theta|^2$. There are a few other empirical regularization techniques for neural networks, such as early stopping (to prevent overfitting) and dropout.

Extended Reading

Durrett, R. (2019). *Probability: theory and examples*, volume 49. Cambridge university press.

Naaman, M. (2021). On the tight constant in the multivariate dvoretzky-kiefer-wolfowitz inequality. *Statistics & Probability Letters*, 173:109088.

Petrov, V. V. (2012). *Sums of independent random variables*, volume 82. Springer Science & Business Media.

Chapter 1

Universal Approximation Theorems

Mathematics is the art of giving the same name to different things.

– Henri Poincaré



HIS chapter will discuss the universal approximation theorems for neural networks. It explains how neural networks of varying widths/depths can fit complex datasets when their parameters are optimized.

1.1 Shallow Neural Network

1.1.1 Qualitative Studies

In this section, we focus on the approximation theorems for the shallow neural networks (also known as single-layer neural networks) with arbitrarily large width.

$$f(x, \theta) = \sum_{i=1}^n c_i \sigma(w_i \cdot x + b_i), \quad x \in X \subset \mathbb{R}^d,$$

where the parameters $\theta = \{(c_i, w_i, b_i) \mid i \in [n]\} \in \mathbb{R}^{(d+2) \times n}$. The function σ is called the **nonlinear** activation function. In addition, we assume X is bounded and compact, e.g., the hypercube $[0, 1]^d$ or the sphere $\mathbb{S}^{d-1}$.

Definition 1.1.1. A signed measure on $(X, \mathcal{A})$ ($\mathcal{A}$ being the Borel σ -algebra) is a function $\mu : \mathcal{A} \rightarrow \mathbb{R} \cup \{\pm\infty\}$ that

1. μ can only attain at most one of $\{-\infty, \infty\}$.
2. $\mu(\emptyset) = 0$.
3. Let $\{E_n\}_{n=1}^\infty \subset \mathcal{A}$ be a sequence of pairwise disjoint sets, then

$$\mu\left(\bigcup_n E_n\right) = \sum_n \mu(E_n).$$

Fact 1.1.2 (Hahn Decomposition). A finite signed measure μ can be written as

$$\mu = \mu^+ - \mu^-$$

where μ^+ and μ^- are two finite non-negative measures.

We then define the Fourier transform on finite signed measures on $X \subset \mathbb{R}^d$, denoted by $\mathcal{M}(X)$.

Theorem 1.1.3. If $\mu \in \mathcal{M}(X)$, then the Fourier transform $\mathcal{F} : \mathcal{M}(X) \rightarrow C_b(\mathbb{R}^d)$

$$\mathcal{F}(\mu) = \hat{\mu}(\zeta) = \int_X e^{-ix \cdot \zeta} \mu(dx)$$

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is well-defined, where C_b means the space of bounded continuous functions.

Proof. Use the Hahn decomposition to define the Fourier transform, then apply the dominated convergence theorem to establish continuity. $\square$

Remark 1.1.4. If $\hat{\mu}_1 = \hat{\mu}_2$, by Fubini's theorem and Fourier transform is invertible on $\mathcal{S}$, for any $\psi \in \mathcal{S}$ ($\mathcal{S}$ means Schwartz function class),

$$\int_X \psi(x) \mu_1(dx) = \int_X \psi(x) \mu_2(dx)$$

Then, μ is orthogonal to $\mathcal{S}$, and $\mathcal{S}$ is dense in $C_0(X)$ (continuous in X , vanish at infinity), by Riesz-Markov theorem, $\mu \in \mathcal{M}(X)$ is unique.

Definition 1.1.5. Let μ be a **finite signed measure** on the data $X \subset \mathcal{X}$, and $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be the activation function. Then σ is called **discriminatory** with respect to μ if

$$\int_X \sigma(w \cdot x + b) \mu(dx) = 0,$$

for every $(w, b) \in \mathbb{R}^{d+1}$ implies $\mu \equiv 0$.

Definition 1.1.6. A continuous function $\sigma : \mathbb{R} \rightarrow [0, 1]$ is called **sigmoidal** if

$$\lim_{x \rightarrow -\infty} \sigma(x) = 0, \quad \lim_{x \rightarrow \infty} \sigma(x) = 1.$$

Example 1.1.7. For instance, the function

$$\sigma(x) = \frac{1}{1 + e^{-x}} = \frac{1}{2} (1 + \tanh \frac{x}{2})$$

is sigmoidal.

Theorem 1.1.8. Sigmoidal functions are discriminatory.

Proof. For sigmoidal functions, the boundedness of σ implies (dominated convergence theorem) that (fix arbitrary w, b, c)

$$\lim_{N \rightarrow \infty} \int_X \sigma(N(w \cdot x + b) + c) \mu(dx) = \mu(\{w \cdot x + b > 0\}) + \sigma(c) \mu(\{w \cdot x + b = 0\}).$$

Therefore (by varying c), for any $(w, b) \in \mathbb{R}^{d+1}$,

$$\mu(\{w \cdot x + b > 0\}) = \mu(\{w \cdot x + b = 0\}) = 0.$$

For any continuous function $h : \mathbb{R} \rightarrow \mathbb{R}$, one can construct a sequence of simple functions that converges to h pointwise,

$$\sum_i a_i \operatorname{sgn}(w \cdot x + b_i) \rightarrow h(w \cdot x).$$

Therefore,

$$\int_X h(w \cdot x) \mu(dx) = 0.$$

It implies $\hat{\mu} = 0$, thus $\mu = 0$ by uniqueness. $\square$

The following theorem is guaranteed by the Riesz-Markov theorem.

Theorem 1.1.9. *If σ is discriminatory, then the shallow neural networks*

$$f(x, \theta) = \sum_{i=1}^N c_i \sigma(w_i \cdot x + b_i)$$

is dense in $C(X)$.

In other words, for any continuous function $h \in C(X)$ and $\varepsilon > 0$, there exists a shallow sigmoidal neural network f that $\|f - h\|_\infty < \varepsilon$.

Remark 1.1.10. *On the other hand, we observe that polynomials of finite degree cannot be discriminatory because polynomials of affine transformation in x are still polynomials.*

Whether a non-polynomial continuous function $\sigma \in C(\mathbb{R})$ is discriminatory? We first consider $\sigma \in C^\infty(\mathbb{R})$. If the signed measure μ is orthogonal to the linear vector space spanned by $\sigma(w \cdot x + b)$, $(w, b) \in \mathbb{R}^{d+1}$. Then μ is orthogonal to

$$\frac{1}{\varepsilon} (\sigma((w + \varepsilon e_i) \cdot x + b) - \sigma(w \cdot x + b))$$

for an arbitrary $\varepsilon > 0$ and $\{e_i\}_{i=1}^d$ denotes the canonical basis in $\mathbb{R}^d$. Thus, μ is orthogonal to linear functions

$$x_i \sigma'(w \cdot x + b)|_{w=0} = x_i \sigma'(b), \quad i \in [d].$$

Similarly, one can take higher derivatives in w along different directions and show

μ is orthogonal to

$$\prod_{k=1}^n x_{i_k} \cdot \sigma^{(n)}(w \cdot x + b)|_{w=0} = \left(\prod_{k=1}^n x_{i_k} \right) \cdot \sigma^{(n)}(b), \quad i_k \in [d].$$

Therefore, μ is orthogonal to all polynomials, which are dense in $C(X)$. We only need the following fact to guarantee all polynomials are included.

Lemma 1.1.11. *If $\sigma \in C^\infty(\mathbb{R})$ is not a polynomial, then for any $n \in \mathbb{N}$, there exists $b_n \in \mathbb{R}$ that $\sigma^{(n)}(b_n) \neq 0$.*

The proof is trivial. Next, we consider $\sigma \in C(\mathbb{R})$ only. We use an arbitrary mollifier $\phi \in C_c^\infty$ to define σ_ϕ as follows.

$$\sigma_\phi(x) := \int_{\mathbb{R}} \sigma(x - y)\phi(y)dy = \int_{\mathbb{R}} \phi(x - y)\sigma(y)dy.$$

Then

$$\sigma_\phi(w \cdot x + b) = \int_{\mathbb{R}} \sigma(w \cdot x + b - y)\phi(y)dy.$$

The linearity implies μ is orthogonal to $\sigma_\phi(w \cdot x + b) \in C^\infty(\mathbb{R})$. Following the previous strategy, we only need to show that all polynomials can be recovered by differentiating σ_ϕ with respect to w .

Otherwise, σ_ϕ must be a polynomial for every ϕ . We consider the highest degree

$$D := \sup_{\phi \in C_c^\infty(\mathbb{R})} \deg(\sigma_\phi).$$

1. If $D = \infty$, then all polynomials can be recovered by taking a maximizing sequence of ϕ .
2. If $D < \infty$, then we take a sequence of ϕ such that $\sigma_\phi \rightarrow \sigma$ uniformly on any compact set, which means σ must be a polynomial itself, contradiction.

This also proves a result in [Hornik et al. \(1989\)](#). The original proof relies on the Stone-Weierstrass theorem because neural networks form an algebra.

Theorem 1.1.12. *The shallow neural networks of the form*

$$f(x, \theta) = \sum_{i=1}^N c_i \prod_{j=1}^{J_i} \sigma(w_{ij} \cdot x + b_{ij})$$

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are dense in $C(X)$.

The previous theorems are under the $C(X)$ topology. Next, we briefly consider the $L^2(X, \nu)$ topology.

Definition 1.1.13. Suppose the function σ satisfies

$$\sigma(w \cdot x + b) \in L^2(X, \nu)$$

for all $(w, b) \in \mathbb{R}^d$. Then σ is **discriminatory** in $L^2(X, \nu)$ if for $f \in L^2(X, \nu)$

$$\int_X \sigma(w \cdot x + b) f(x) \nu(dx) = 0$$

for all $(w, b) \in \mathbb{R}^{d+1}$ implies $f = 0$.

The basic Riesz representation theorem applies here because $L^2(X, \nu)$ is a Hilbert space.

Theorem 1.1.14. If σ is discriminatory in $L^2(X, \nu)$, then the shallow neural networks

$$f(x, \theta) = \sum_{i=1}^N c_i \sigma(w_i \cdot x + b_i)$$

are dense in $L^2(X, \nu)$.

Proof. The span by $f(x, \theta)$ forms a linear subspace $V \subset L^2(X, \nu)$. If V is not dense in $L^2(X, \nu)$, there is a linear functional $\psi : L^2(X, \nu) \rightarrow \mathbb{R}$ such that $\psi(V) = 0$ and $\psi \neq 0$ otherwise. The Riesz representation theorem implies

$$\psi(h) = \int_X g_\psi(x) h(x) \nu(dx).$$

for some nonzero $g_\psi \in (L^2(X, \nu))^* = L^2(X, \nu)$ (reflective). Then by definition,

$$\int_X g_\psi(x) \sigma(w \cdot x + b) \nu(dx) = 0$$

for all $(w, b) \in \mathbb{R}^{d+1}$. Therefore, $g_\psi = 0$, contradiction. $\square$

Example 1.1.15. *The indicator function of half-planes*

$$\sigma(w \cdot x + b) = \text{sgn}(w \cdot x + b)$$

are discriminatory in $L^2(X, \nu)$ (since they are dense in stronger topology).

Example 1.1.16. *The functions $\sin(w \cdot x + b)$ are discriminatory in $L^2(X, \nu)$.*

1.1.2 Quantitative Studies

In this section, we quantify the approximation error of the shallow neural network in the simplest setting. Regarding the approximation error, let h be a function in a specific set $\mathcal{H}$, and the approximation error means we would like to find out the error as follows.

$$\inf_{\theta} \left\| h - \sum_{i=1}^n c_i \sigma(w_i \cdot x + b_i) \right\|. \quad (1.1)$$

Common choices for the norm $\|\cdot\|$ can be $C(X)$, $L^2(X)$, etc.

Remark 1.1.17. *Note that for a different $h \in \mathcal{H}$, it allows different parameters θ . Consider another type of approximation error as follows.*

$$\inf_{(w_i, b_i)} \sup_{h \in \mathcal{H}} \inf_{c_i} \left\| h - \sum_{i=1}^n c_i \sigma(w_i \cdot x + b_i) \right\|. \quad (1.2)$$

Then, the approximation error (1.1) is no less than (1.2). The error (1.2) is similar to the Kolmogorov n -width.

In general, the rougher the objective function, the more difficult it is to approximate with activation functions. We first introduce a classical result by Andrew Barron, as presented in Barron (1993, 1994). For simplicity, we assume $X = B_1(\mathbf{0})$ as the unit ball centered at the origin.

Definition 1.1.18 (Barron Space). *The Barron space $\mathcal{B}(X)$ is defined by*

$$\mathcal{B}(X) = \left\{ f \in L^1(X) \mid \int_{\mathbb{R}^d} \|\zeta\|_2 |\widehat{f}(\zeta)| d\zeta < \infty \right\},$$

where $\widehat{f}$ denotes the Fourier transform of f . We set

$$\|f\|_{\mathcal{B}} = \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} \|\zeta\|_2 |\widehat{f}(\zeta)| d\zeta$$

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as the norm for $f \in \mathcal{B}(X)$.

Question 1.1.19. For any $k > \frac{1}{2}(d+1)$,

$$W^{k,2}(X) \subset \mathcal{B}(X) \subset W^{1,\infty}(X).$$

Definition 1.1.20. Let $K \subset L^2(X)$ be a bounded set, and let $\text{co}(K)$ denote the **convex hull** of K .

$$\text{co}(K) = \left\{ \sum_i a_i k_i \mid k_i \in K, a_i \geq 0, \sum_i a_i \leq 1 \right\}.$$

Definition 1.1.21. We consider a covering of K by n balls in the L^2 metric. The optimal diameter is defined as follows.

$$R_n(K) := \inf\{r > 0 \mid K \text{ can be covered by } n \text{ balls of diameter } r\}.$$

Theorem 1.1.22 (Makovoz (1996)). Let $f \in \text{co}(K)$. There exists a function $\mathcal{G}^*$ in the form of

$$\mathcal{G}^* = \sum_{j=1}^{2n} a_j k_j, \quad a_j \geq 0, \quad \sum_j a_j \leq 1,$$

such that

$$\|f - \mathcal{G}^*\|_{L^2(X)} \leq \frac{R_n(K) + \varepsilon}{\sqrt{n}},$$

for an arbitrarily small $\varepsilon > 0$.

Proof. If $f \in \text{co}(K)$, then there exists $k_i \in K, i = 1, \dots, m$ such that

$$f = \sum_{i=1}^m a_i k_i, \quad a_i \in (0, 1], \quad \sum_{i=1}^m a_i \leq 1,$$

The set $\{k_i\}_{i=1}^m$ can be divided into n groups such that each group is contained

in a ball of radius $R_n(K) + \varepsilon$ for any $\varepsilon > 0$. Without loss of generality, we can assume $m > n$ and each group is non-empty.

Let those balls be $B_1, \dots, B_n$, there is a surjective mapping $\pi : [m] \rightarrow [n]$ such that $k_i \in B_{\pi(i)}$. Then we construct a random variable $\mathcal{G}$ as follows.

$$\mathcal{G} = \sum_{j=1}^n \mathcal{G}_j,$$

where each $\mathcal{G}_j$ is an independent random variable in the following summation form,

$$\mathcal{G}_j = \frac{A_j}{n_j} \sum_{l=1}^{n_j} g_{j,l}, \quad A_j = \sum_{i, \pi(i)=j} a_i, \quad n_j = \lfloor nA_j \rfloor + 1,$$

where $g_{j,l}$ of the same j are i.i.d. random variables provided that $\pi(l) = j$.

$$\mathbb{P}[g_{j,l} = k_i] = \frac{a_i}{A_j}.$$

Therefore, $\mathbb{E}[\mathcal{G}_j] = f_j := \sum_{i, \pi(i)=j} a_i k_i$. Now, we compute

$$\begin{aligned} \mathbb{E}[\|\mathcal{G} - f\|^2] &= \sum_{j=1}^n \mathbb{E}[\|\mathcal{G}_j - f_j\|^2] \\ &= \sum_{j=1}^n \left(\frac{A_j}{n_j} \right)^2 n_j \mathbb{E} \left[\left\| g_{j,l} - \frac{f_j}{A_j} \right\|^2 \right] \\ &\leq \frac{(R_n(K) + \varepsilon)^2}{n} \sum_{j=1}^n A_j \leq \frac{(R_n(K) + \varepsilon)^2}{n}. \end{aligned}$$

It means there exists a realization $\mathcal{G}^*$ consists of at most $\sum_{j=1}^n n_j \leq 2n$ terms from K that

$$\|\mathcal{G}^* - f\|_{L^2(X)} \leq \frac{(R_n(K) + \varepsilon)}{\sqrt{n}},$$

and ε can be arbitrarily small. □

Remark 1.1.23. The result holds if $f \in \overline{\text{co}(K)}$, since one can find a sequence $f_j \in \text{co}(K)$ arbitrarily close to f .

Remark 1.1.24. In [Barron \(1993\)](#), the set K is treated as a single group, then the form of $2n$ terms can be replaced by n terms, but the $R_n(K)$ is replaced by $R_1(K)$. Therefore, the above result is better.

Remark 1.1.25. The key idea is to treat f as an expectation with respect to a probability measure on K . Then, the Monte Carlo error $\mathcal{O}(n^{-1/2})$ is guaranteed for a single

realization.

Next, we prove the approximation error bound for Barron space.

Theorem 1.1.26. *Let $\mathcal{B}_{1/2} = \{f \in \mathcal{B}(X) \mid \|f\|_{\mathcal{B}} \leq \frac{1}{2}\}$. Suppose σ is a sigmoidal function, then $\forall f \in \mathcal{B}_{1/2}$, there exists a shallow neural network (with a constant shift)*

$$g = \sum_{i=1}^{2n} a_i \sigma(w_i \cdot x + b_i) + f(0)$$

that

$$\|f - g\|_{L^2(X)} \leq C n^{-\frac{d+1}{2d}} + \varepsilon,$$

where C is an absolute constant and $\varepsilon > 0$ is arbitrary.

Proof. Let K be the bounded set in $L^2(X)$:

$$K = \left\{ \pm \sigma(w \cdot x + b) \mid w \in \mathbb{R}^d, b \in \mathbb{R} \right\}.$$

When $\lambda \rightarrow \infty$, the limit $\sigma(\lambda(w \cdot x + b))$ becomes the Heaviside function σ_h with the jump at $w \cdot x + b = 0$. Define

$$K_h = \left\{ \pm \sigma_h(w \cdot x + b) \mid w \in \mathbb{R}^d, b \in \mathbb{R} \right\}.$$

Then, $\overline{\text{co}(K_h)} \subset \overline{\text{co}(K)}$. Due to the Heaviside function's affinity and boundedness of X , we can restrict w and b to a smaller set,

$$K_h = \left\{ \pm \sigma_h(w \cdot x + b) \mid w \in \mathbb{R}^d, \|w\| = 1, |b| \leq 1 \right\}.$$

For a fixed $w \in \mathbb{R}^d$, our first observation is that

$$\{g(w \cdot x) \mid g : [-\|w\|_2, \|w\|_2] \rightarrow \mathbb{R}, g(0) = 0, \|g\|_{TV} \leq 1\} \subset \overline{\text{co}(K_h)},$$

where $\|g\|_{TV}$ is the total variation. Therefore, $\forall w, b \in \mathbb{R}^d \times \mathbb{R}$ that $w \neq \mathbf{0}$,

$$\frac{1}{2\|w\|_2} (\cos(w \cdot x + b) - \cos(b)) \in \overline{\text{co}(K_h)}, \quad x \in X. \quad (1.3)$$

Because

$$\begin{aligned}
f(x) - f(0) &= \Re \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} (e^{iw \cdot x} - 1) \widehat{f}(w) dw \\
&= \Re \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} (e^{iw \cdot x} - 1) e^{i\theta(w)} |\widehat{f}(w)| dw \\
&= \frac{1}{(2\pi)^{d/2}} \int_{\mathbb{R}^d} (\cos(w \cdot x + \theta(w)) - \cos(\theta(w))) |\widehat{f}(w)| dw \\
&= \int_{\mathbb{R}^d} \left[\frac{1}{2\|w\|_2} (\cos(w \cdot x + \theta(w)) - \cos(\theta(w))) \right] \frac{2\|w\|_2}{(2\pi)^{d/2}} |\widehat{f}(w)| dw,
\end{aligned}$$

where $\theta(w) = \arg \widehat{f}(w)$. Meanwhile, we notice that

$$\int_{\mathbb{R}^d} \frac{2\|w\|_2}{(2\pi)^{d/2}} |\widehat{f}(w)| dw = 2\|f\|_{\mathcal{B}} \leq 1,$$

therefore, $f(x) - f(0) \in \overline{\text{co}(K_h)}$.

Finally, we estimate $R_n(K_h)$. When $\|w - w'\| \leq \varepsilon^2$ and $|b - b'| \leq \varepsilon^2$,

$$\int_X |\sigma_h(w \cdot x + b) - \sigma_h(w' \cdot x + b')|^2 dx \leq C\varepsilon^2$$

for a certain absolute constant C . Therefore, $R_n(K) \leq C'n^{-\frac{1}{2d}}$ for another absolute constant C' . Then, applying the previous Theorem of Makovoz yields the desired approximation error bound. $\square$

Remark 1.1.27. All arguments can be adapted to a general $L^2(X, \mu)$, which is equipped with a general measure. In fact, we only need a Hilbert space. Thus, the approximation error can be calculated by the empirical loss.

Remark 1.1.28. The constant shift can be constructed by an additional term $\sigma(w \cdot x + b)$ with $b \rightarrow \infty$. Thus, the conclusion holds if the shallow neural network g consists of $2n + 1$ terms.

Remark 1.1.29. The error bound of the above theorem is slightly better than that of the original bound in Barron (1993), where the error is $\mathcal{O}(n^{-1/2})$. We should also notice that when $d = 1$, the bound $\mathcal{O}(n^{-1})$ is optimal.

Example 1.1.30. Let $f(x) = x$ on $[-1, 1]$. Let g be a sum of n Heaviside functions with jumps at $x_0, \dots, x_{n+1}$ (including the endpoints of the domain), then

$$\|f - g\|^2 = \sum_{i=0}^n \int_{x_i}^{x_{i+1}} |x - c_i|^2 dx \leq \frac{1}{12} \sum_{i=0}^n |x_{i+1} - x_i|^3 \leq \frac{2}{3} \frac{1}{(n+1)^2}.$$

This means the optimal approximation error for f with a sum of n Heaviside functions is of the order of $\mathcal{O}(n^{-1})$.

Question 1.1.31. Is the bound $\mathcal{O}(n^{-\frac{d+1}{2d}})$ optimal for high dimensions? It is answered positively in [Siegel and Xu \(2024\)](#).

More importantly, the error bound in [Barron \(1993\)](#) shows no dependence on the dimension of the domain X . It initiated an unfortunate discussion about **defeating the curse of dimensionality**.

Remark 1.1.32. The curse of dimensionality refers to the general phenomenon in which high dimensionality makes the problem substantially more difficult. For approximation theory, it usually means the approximation error roughly scales like

$$(\text{number of terms})^{-\frac{\text{regularity}}{\text{space dimension}}}$$

This upper bound usually can be constructed.

Example 1.1.33. We consider function $f(x) = f(|x|) = (1 + |x|^2)^{-d/2}$, its Fourier transform is the Bessel kernel

$$\widehat{f}(\zeta) = \frac{2\pi^{d/2}}{\underbrace{\Gamma(d/2)}_{=|\mathbb{S}^{d-1}|}} K_0(|\zeta|).$$

Then the Barron norm of f grows exponentially fast in dimension.

$$\begin{aligned} \|f\|_{\mathcal{B}} &= \frac{1}{(2\pi)^{d/2}} \left(\frac{2\pi^{d/2}}{\Gamma(d/2)} \right)^2 \int_0^\infty r^d K_0(r) dr \\ &= \frac{1}{(2\pi)^{d/2}} \left(\frac{2\pi^{d/2}}{\Gamma(d/2)} \right)^2 2^{d-1} \left(\Gamma\left(\frac{d+1}{2}\right) \right)^2 \\ &= (2\pi)^{d/2} \left(\frac{\Gamma((d+1)/2)}{\Gamma(d/2)} \right)^2. \end{aligned}$$

It means in high dimensions, the Barron space excludes many smooth functions.

Here, we have used the identity ([DLMF, 2025, \(10.43.19\)](#)):

$$\int_0^\infty x^{s-1} K_t(x) dx = 2^{s-2} \Gamma\left(\frac{s+t}{2}\right) \Gamma\left(\frac{s-t}{2}\right), \quad t > s.$$

Question 1.1.34. Explain why the curse of dimensionality is NOT defeated in Barron's result.

We note that the probabilistic error bound depends on the prior knowledge of the sampling distribution, while the distribution is inherently high-dimensional. The result establishes only the existence of a shallow neural network, without providing a computationally tractable algorithm. Jones, see Jones (1992), proposes a greedy algorithm with n steps to construct the shallow neural network iteratively as follows.

$$(\alpha', g') = \arg \min_{\alpha \in [0,1], g \in \overline{K(\sigma)}} \|f - (\alpha h_{s-1} + (1 - \alpha)g)\|,$$

and generate $h_s = \alpha' h_{s-1} + (1 - \alpha')g'$. The algorithm can achieve the desired $\mathcal{O}(n^{-1/2})$ bound, but its search space is large (depends on the dimension).

Question 1.1.35. Prove this algorithm can achieve the error bound $\mathcal{O}(n^{-1/2})$ after n steps for $f \in \mathcal{B}_{1/2}$.

1.1.3 Ridge Bounds

In Barron (1993), the approximation theory is specifically for Barron space and the Heaviside activation function σ_h . What if we allow arbitrary activation functions in the Sobolev space? The following set

$$\mathcal{R}_n = \left\{ \sum_{i=1}^n k_i(w_i \cdot x) \mid k_i \in C(\mathbb{R}), w_i \in \mathbb{R}^d, i \in [n] \right\}$$

consists of all n -term **ridge** functions. The following two theorems of Maierov are optimal. The exponent exhibits the curse of dimensionality.

Theorem 1.1.36 (Maierov (1999)). Let $m \geq 1, d \geq 2$, and X denotes the unit ball in $\mathbb{R}^d$. There exists a function $f \in W^{m,2}(X)$ that $\|f\|_{W^{m,2}} \leq 1$ and

$$\inf_{g \in \mathcal{R}_n} \|f - g\|_{L^2(X)} \geq C n^{-\frac{m}{d-1}},$$

where C is independent of f and n .

The proof for the lower bound is technical. Interested readers can read [Maiorov \(1999\)](#) for details.

Theorem 1.1.37 (Maiorov (1999)). Let $m \geq 1$, $d \geq 2$, and X denotes the unit ball in $\mathbb{R}^d$. For any $f \in W^{m,2}(X)$ that $\|f\|_{W^{m,2}} \leq 1$,

$$\inf_{g \in \mathcal{R}_n} \|f - g\|_{L^2(X)} \leq C n^{-\frac{m}{d-1}},$$

where C is independent of f and n .

☕ Preparations for Upper Bound

Before the proof of the upper bound, we state a classical theorem for approximation theory.

Theorem 1.1.38 (Canuto and Quarteroni (1982)). Let $f \in W^{m,2}(X)$ that $\|f\|_{W^{m,2}} \leq 1$. Denote $\mathcal{P}_k$ the linear space of polynomials of degree $\leq k$ in $\mathbb{R}^d$. Then

$$\inf_{g \in \mathcal{P}_k} \|f - g\|_{L^2(X)} \leq C k^{-m},$$

where C is independent of k and f .

Proof. The extension theorem of Stein allows us to consider X as a cube $[-1, 1]^d$ instead of a ball. Then on X , the orthogonal polynomials are

$$Q_{l_1, \dots, l_d}(x) = \prod_{i=1}^d P_{l_i}(x_i),$$

where P_{l_i} is the normalized Legendre polynomial of order $l_i \in \mathbb{N}$. Each P_{l_i} satisfies

$$\mathcal{A}_i P_{l_i}(x_i) = -l_i(l_i + 1)P_{l_i}, \quad \mathcal{A}_i := \partial_i(1 - x_i^2)\partial_i.$$

The operator $\mathcal{A}_i$ is a self-adjoint operator. Therefore, let $p = m/2$ (assume m is even),

$$(-l_i(l_i + 1))^p \langle f, Q_{l_1, \dots, l_d} \rangle = \langle f, \mathcal{A}_i^p Q_{l_1, \dots, l_d} \rangle = \langle \mathcal{A}_i^p f, Q_{l_1, \dots, l_d} \rangle.$$

The approximation error satisfies

$$\begin{aligned}
\inf_{g \in \mathcal{P}_k} \|f - g\|_{L^2(X)}^2 &= \sum_{l_1 + \dots + l_d > k} |\langle f, Q_{l_1, \dots, l_d} \rangle|^2 \\
&\leq \sum_{i=1}^d \sum_{l_i > k/d} |\langle f, Q_{l_1, \dots, l_d} \rangle|^2 \\
&= \sum_{i=1}^d \sum_{l_i > k/d} \left(\frac{1}{l_i(l_i + 1)} \right)^{2p} |\langle \mathcal{A}_i^p f, Q_{l_1, \dots, l_d} \rangle|^2 \\
&\leq \left(\frac{d}{k} \right)^{4p} \sum_{i=1}^d \|\mathcal{A}_i^p f\|_{L^2(X)}^2.
\end{aligned}$$

It remains to show $\|\mathcal{A}_i f\|_{L^2(X)} \leq C\|f\|_{W^{2,2}(X)}$, which is straightforward. $\square$

Remark 1.1.39. If m is an odd integer, we can use $(-\mathcal{A}_i)^{1/2}$, which is uniquely determined by the eigendecomposition. And

$$\langle (-\mathcal{A}_i)^{1/2} f, (-\mathcal{A}_i)^{1/2} f \rangle = \langle -\mathcal{A}_i f, f \rangle \leq \|f\|_{W^{1,2}(X)}^2.$$

Remark 1.1.40. For orthogonal polynomials on the unit ball, they are also eigenfunctions of a Sturm-Liouville type differential operator. Similar results still hold.

Question 1.1.41. Let $f \in W^{m,2}(\mathbb{R}^d, \mu)$ with $\mu(dx) = e^{-x^2} dx$ that

$$\|f\|_{W^{m,2}(\mathbb{R}^d, \mu)} \leq 1.$$

Prove

$$\inf_{g \in \mathcal{P}_k} \|f - g\|_{L^2(\mathbb{R}^d, \mu)} \leq Ck^{-m/2}.$$

Remark 1.1.42. The discounted performance for weighted Sobolev spaces on $\mathbb{R}^d$ is not surprising since the constant C is proportional to $(\text{diam}(X))^m$. The n th Hermite polynomials (and Laguerre polynomials as well) are eigenfunctions of eigenvalues $\mathcal{O}(n)$. We should expect a similar error bound (see [Guo et al. \(2005\)](#)) for $W^{m,2}(\mathbb{R}_+^d, \nu)$ that

$$\|f\|_{W^{m,2}(\mathbb{R}_+^d, \nu)}^2 = \sum_{|\alpha| \leq m} \int_{\mathbb{R}_+^d} |f^{(\alpha)}(x)|^2 x^\alpha e^{-|x|_1} dx.$$

☕ Proof for Upper Bound

The proof uses the duality of differential operators ∂^α to polynomial x^α , where α is a multi-index.

Proof of Theorem 1.1.37. We construct $\mathcal{P}_k$ as the linear space of homogeneous polynomials of degree k in $\mathbb{R}^d$. Then let $n = \dim(\mathcal{P}_k) = \binom{d-1+k}{k} \leq Ck^{d-1}$, where C depends on d , there exists (why?) $w_i \in \mathbb{R}^d, i \in [n]$ that

$$\bigcup_{l=0}^k \mathcal{P}_l = \left\{ \sum_{i=1}^n k_i(w_i \cdot x) \mid k_i \in \bigcup_{l=0}^k \pi_l \right\} \subset \mathcal{R}_n,$$

where π_l denotes the linear space of homogeneous polynomials of degree k in $\mathbb{R}^1$. Thus,

$$\inf_{g \in \mathcal{R}_n} \|f - g\|_{L^2(X)} \leq \inf_{g \in \bigcup_{l=0}^k \mathcal{P}_l} \|f - g\|_{L^2(X)} \leq C' k^{-m} \leq C'' n^{-\frac{m}{d-1}}.$$

□

Question 1.1.43. Complete the above proof.

Question 1.1.44. Since the Barron space $\mathcal{B} \supset W^{m,2}$ for $m > \frac{d+1}{2}$, compare the error bounds from the previous section and Maiorov's result.

The result of Maiorov shows that the approximation error of Ridge functions is bounded by $\mathcal{O}(n^{-m/(d-1)})$ for the Sobolev space $W^{m,2}$. Thus, shallow neural networks of width n with any activation function must inherit the lower bound. But how sharp is the upper bound?

Theorem 1.1.45 (Maiorov and Pinkus (1999)). There exists $\sigma \in C^\infty$, sigmoidal, strictly increasing, and for any $g \in \mathcal{R}_n$ and $\varepsilon > 0$, there are parameters (c_i, w_i, b_i) that

$$\left| g(x) - \sum_{i=1}^{n+d+1} c_i \sigma(w_i \cdot x + b_i) \right| < \varepsilon, \quad x \in X.$$

Rough Idea. The trick is to construct a function by joining a countable dense set of continuous functions with disjoint support. We give the main steps.

1. $C[-1, 1]$ is separable, so there is a countable dense subset

$$\{u_k\} \subset C^\infty[-1, 1],$$

For instance, the polynomial ring $\mathbb{Q}[x]$ by Weierstrass theorem.

2. On the real axis, put the function $\sigma = a_k u_k + b_k \phi + c_k \psi$ on the disjoint intervals I_k , where ϕ and ψ are linear functions supported on two intervals.
3. Choose the parameters a_k, b_k, c_k such that the whole function on $\mathbb{R}$ is increasing. Then fill the gaps between the intervals smoothly.

Then for any function $g \in \mathcal{R}_n$, it consists of n ridge functions $g_i, i = 1, \dots, n$. Each g_i is made with σ on an interval and two linear functions in $\mathbb{R}^d$. However, there are at most $(d+1)$ linear functions. Therefore, at most we need $\min(3n, n + d + 1)$ terms. $\square$

Question 1.1.46. Complete this proof.

Since the dimension d is fixed, $n \sim n + d + 1$. This means that the activation function σ in the above theorem is sharp for approximation in Sobolev spaces.

Corollary 1.1.47. Let σ be the activation function in the previous theorem, and define

$$\mathcal{S} := \left\{ \sum_{i=1}^n c_i \sigma(w_i \cdot x + b_i) \mid c_i \in \mathbb{R}, w_i \in \mathbb{R}^d, b_i \in \mathbb{R} \right\}.$$

Then, for any $f \in W^{m,2}(X)$ with bound $\|f\|_{W^{m,2}(X)} \leq 1$,

$$\inf_{g \in \mathcal{S}} \|f - g\|_{L^2(X)} \leq C n^{-m/(d-1)}.$$

However, such an activation function is too complex (even pathological) to be practical, despite its desirable properties, including smoothness (even analyticity when $n + d + 1$ is replaced by $3n$), monotonicity, and sigmoidality. Computationally, simple activation functions are preferred. In return, the approximation error bound could be slightly suboptimal.

☕ Stable Approximation

In this section, we show that the error bound $\mathcal{O}(n^{-m/d})$ is optimal when the approximation is a *continuous* mapping. In this case, we say it is a *stable* approximation. Note that this is *not* about the representation itself; it is related to how the approximation is found.

Definition 1.1.48 (Nonlinear n -Width). Let K be a compact set in a normed space $\mathcal{X}$. Let $\mathcal{E}_n : K \rightarrow \mathbb{R}^n$ be a continuous map and $\mathcal{D}_n : \mathbb{R}^n \rightarrow K$ be a function, unnecessarily continuous. Define

$$h_n(K; \mathcal{X}) = \inf_{\mathcal{E}_n, \mathcal{D}_n} \sup_{f \in K} \|f - \mathcal{D}_n(\mathcal{E}_n(f))\|_{\mathcal{X}},$$

$h_n(K)$ is the continuous nonlinear n -width.

The following theorem is proved in DeVore et al. (1989).

Theorem 1.1.49 (DeVore et al. (1989)). Let $m, p, d \geq 1$, and X be the unit ball in $\mathbb{R}^d$. Then

$$h_n(W^{m,p}(X); L^p(X)) \asymp n^{-m/d}.$$

Proof. The upper bound is not hard, as we can see later. We only show the lower bound. The purpose is to introduce the standard approach.

1. Set (Bernstein n -width)

$$b_n(K; \mathcal{X}) := \sup_{\mathcal{X}_{n+1}} \sup\{\lambda \mid \lambda B(\mathcal{X}_{n+1}) \subset K\},$$

where $\mathcal{X}_{n+1} \subset \mathcal{X}$ is any $(n+1)$ dimensional subspace of $\mathcal{X}$, $B(\mathcal{X}_{n+1})$ is the unit ball in $\mathcal{X}_{n+1}$ (metric in $\mathcal{X}$).

2. Let $T : K \rightarrow \mathbb{R}^n$ that $T(f) = \mathcal{E}_n(f) - \mathcal{E}_n(-f)$, then T is odd on $\lambda \partial B(\mathcal{X}_{n+1})$, which means there exists $f^* \in \partial \lambda B(\mathcal{X}_{n+1})$ that $T(f^*) = 0$ because of the famous *Borsuk-Ulam* theorem. Thus,

$$2f^* = (f^* - \mathcal{D}_n(\mathcal{E}_n(f^*))) - (-f^* - \mathcal{D}_n(\mathcal{E}_n(-f^*))).$$

Hence (by pigeonhole principle), there exists $f^\dagger = f^*$ (or $-f^*$) that

$$\|f^\dagger - \mathcal{D}_n(\mathcal{E}_n(f^\dagger))\|_{\mathcal{X}} \geq \lambda = b_n(K; \mathcal{X}).$$

This holds for any $\mathcal{E}_n$ and $\mathcal{D}_n$, thus $h_n(K; \mathcal{X}) \geq b_n(K; \mathcal{X})$.

3. We see that $b_n(K; \mathcal{X})$ is independent of $\mathcal{E}_n$ and $\mathcal{D}_n$. The proof is standard by construction.

Let $K = \{f \in W^{m,p}(X) \mid \|f\|_{W^{m,p}} \leq 1\}$, which is compact in $\mathcal{X} = L^p$. Take ϕ as a $C_c^\infty[0, 1]^d$ function (e.g., standard bump function). Decompose the domain $X \subset \mathbb{R}^d$ into disjoint small cells of size ℓ and shift/rescale ϕ to each cell (denoted by ϕ_i^*), then it automatically defines $n = \mathcal{O}(\ell^{-d})$ dimensional subspace of $W^{m,p}(X)$ through

$$\mathcal{X}_{n+1} = \left\{ \sum_i c_i \phi_i^* \mid c_i \in \mathbb{R} \right\}.$$

For any $f \in \partial B(\mathcal{X}_{n+1})$, it satisfies that

$$\|f\|_{L^p}^p = \left(\sum_i |c_i|^p \right) \|\phi\|_{L^p}^p = 1.$$

and we need to find the largest λ that

$$\|\lambda f\|_{W^{m,p}}^p = \lambda^p \sum_{|\alpha| \leq m} \left(\ell^{-p|\alpha|} \sum_i |c_i|^p \right) \|\partial^\alpha \phi\|_{L^p}^p \leq 1.$$

Therefore, $\lambda \asymp \ell^m \asymp n^{-m/d}$.

□

Corollary 1.1.50. Let $\mathcal{A}_n$ be an algorithm (or map) to compute the parameterization θ of a shallow neural network of width n for an input function $f \in L^2(X)$. If $\mathcal{A}$ is **continuous**, then there exists $f \in W^{m,2}(X)$ that $\|f\|_{W^{m,2}} \leq 1$ and

$$\|f - \mathcal{A}_n f\|_{L^2} \geq C n^{-m/d}.$$

Remark 1.1.51. Therefore, the optimal error bound $\mathcal{O}(n^{-m/(d-1)})$ that appears in n -term shallow neural network (see Corollary 1.1.47) cannot be implemented by a **continuous** algorithm.

Remark 1.1.52. Iterative algorithms do not mean they are continuous.

☕ Error Bound for General Activations

The following theorem implies that a general activation function can achieve the (stable approximation) error of $\mathcal{O}(n^{-m/d})$ for Sobolev spaces.

Theorem 1.1.53 (Mhaskar (1996)). Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ and $\sigma \in C^\infty(I)$ on some interval I but not a polynomial on I .

$$\mathcal{S} := \left\{ \sum_{i=1}^n c_i \sigma(w_i \cdot x + b_i) \mid c_i \in \mathbb{R}, w_i \in \mathbb{R}^d, b_i \in \mathbb{R} \right\}.$$

Then, for any $f \in W^{m,2}(X)$ with bound $\|f\|_{W^{m,2}(X)} \leq 1$,

$$\inf_{g \in \mathcal{S}} \|f - g\|_{L^2(X)} \leq C n^{-m/d}.$$

Rough Idea. Using Theorem 1.1.37 (and its proof in Section 1.1.3), the set

$$\bigcup_{l=0}^k \mathcal{P}_l = \left\{ \sum_{i=1}^s k_i(w_i \cdot x) \mid k_i \in \bigcup_{l=0}^k \pi_l \right\}$$

provides an approximation error of $\mathcal{O}(k^{-m})$, where $s \approx \mathcal{O}(k^{d-1})$. Each $k_i(w_i \cdot x)$ has $\deg(k_i) \leq k$, and can be represented by a limiting finite difference scheme of $k+1$ terms (why?)

$$k_i(w_i \cdot x) = \lim_{\varepsilon \rightarrow 0} \sum_{j=1}^{k+1} c_{i,j,\varepsilon} \sigma(w_i \cdot x + b_{i,j,\varepsilon}).$$

Therefore,

$$\bigcup_{l=0}^k \mathcal{P}_l \subset \overline{\left\{ \sum_{i=1}^{s(k+1)} c_i \sigma(w_i \cdot x + b_i) \mid c_i, b_i \in \mathbb{R}, w_i \in \mathbb{R}^d \right\}}.$$

Therefore, the total number of terms is $\mathcal{O}(k^d)$, meaning the approximation error is $\mathcal{O}(n^{-m/d})$. $\square$

Question 1.1.54. Complete this proof.

Remark 1.1.55. This theorem shows that a shallow neural network can first approxi-

mate the polynomial basis in $\mathcal{P}_k$ and then combine the basis elements to achieve $\mathcal{O}(n^{-m/d})$ error. Note that this process is entirely linear and thus depends continuously on the function to be approximated.

1.1.4 ReLU Bounds

We now consider specific activation functions within the ReLU class. The ReLU (rectified linear unit) is a common activation function defined by

$$\sigma(x) = \max(0, x). \quad (1.4)$$

More generally, we denote $\sigma^k(x) = (\max(0, x))^k$, then $k = 0$ means the Heaviside function. Since the ReLU class is affine, we only consider $w \in \mathbb{S}^{d-1}$ and $|b| \leq 2 \operatorname{diam}(X)$. Notice that the ReLU class does not satisfy the requirements in Theorem 1.1.53.

Remark 1.1.56. In 1D, the ReLU class is precisely the set of spline functions. The results are classical from finite elements. Let

$$\mathcal{S} = \left\{ \sum_{i=1}^n c_i \sigma^k(x + b_i) \mid b_i, c_i \in \mathbb{R} \right\},$$

then for $1 \leq m \leq k + 1$,

$$\inf_{g \in \mathcal{S}} \|f - g\|_{L^2} \leq C n^{-m} \|f\|_{W^{m,2}}.$$

This is different from Theorem 1.1.53, which allows all m . Since $g^{(m)} \equiv 0$ for higher $m \geq k + 1$, which means $g \in \mathcal{S}$ cannot affect higher derivatives.

In Petrushev (1998), it shows that the ridge approximation error in $\mathbb{R}^1$ can imply the neural network's approximation error in $\mathbb{R}^d$.

Theorem 1.1.57 (Petrushev (1998)). Suppose $1 \leq m \leq \frac{2k+d+1}{2}$, and $f \in W^{m,2}(X)$ and $\|f\|_{W^{m,2}(X)} \leq 1$. Let

$$\mathcal{S} = \left\{ \sum_{i=1}^n c_i \sigma^k(w_i \cdot x + b_i) \mid w_i \in \mathbb{S}^{d-1}, b_i, c_i \in \mathbb{R} \right\}.$$

Then

$$\inf_{g \in \mathcal{S}} \|f - g\|_{L^2(X)} \leq C n^{-m/d}.$$

Remark 1.1.58. In addition to the classical results in Sobolev spaces, we can consider using σ^k as the activation function, and $f(x) = \mathbb{E}_{(w,b) \sim \Theta} [\sigma^k(w \cdot x + b)]$, as discussed at the beginning of this chapter. The result for $k = 0$ is known by Theorem 1.1.22 and Theorem 1.1.26, the L^2 approximation bound is $\mathcal{O}(n^{-\frac{1}{2} - \frac{1}{2d}})$.

In conclusion, shallow neural networks already suffer from the **curse of dimensionality** due to the nature of approximation error, making it impractical to approximate general functions in high dimensions with them. In the next section, we will show that multi-layer neural networks can overcome the limitations of representation (without defeating the curse of dimensionality).

1.2 Multi-Layer Perceptron

We first focus on the two-hidden-layer MLP model. The main difference between MLPs and shallow neural networks is that the additional layer provides substantially greater flexibility for approximation when we do not impose strict requirements on the activation function.

Indeed, the shallow neural network structure intrinsically has difficulty representing locally supported functions in $\mathbb{R}^d$, $d \geq 2$. However, with an additional activation layer, the MLP can approximate locally supported functions.

Question 1.2.1. Try to construct a function

$$\sigma_h \left(\sum_{i=1}^n c_i \sigma_h(w_i \cdot x + b_i) \right)$$

to be compactly supported in the unit cube, where σ_h is the Heaviside function.

1.2.1 Kolmogorov Superposition Theorem

The KST yields a negative answer to Hilbert's 13th problem, which is one of the 23 Hilbert problems compiled in 1900 by David Hilbert.

Problem 1.2.2 (Hilbert's 13th Problem). Prove whether a solution exists for all 7th-degree equations using continuous functions of two arguments.

The KST theorem is also called the *Kolmogorov-Arnold Representation theorem*, which states that every multivariate continuous function $f : [0, 1]^n \rightarrow \mathbb{R}$ can be **represented** as a superposition of continuous single-variable functions.

Theorem 1.2.3 (KST, Original). For any continuous function $f : [0, 1]^d \rightarrow \mathbb{R}$, there exists continuous functions $\phi_{ij} : [0, 1] \rightarrow \mathbb{R}$, and $\Psi_j \in C[0, 1]$, $i \in [n]$, $j \in [2d + 1]$, such that

$$f(x_1, \dots, x_d) = \sum_{j=1}^{2d+1} \Psi_j \left(\sum_{i=1}^d \phi_{ij}(x_i) \right). \quad (1.5)$$

The functions ϕ_{ij} are universal (independent of f).

It can be shown that the outer functions Ψ_j can be shared. The inner functions can also be shared if shifts are allowed. The inner functions can be made Lipschitz continuous, but the outer function cannot be more regular.

Theorem 1.2.4 (KST, Variant). For any continuous function $f : [0, 1]^d \rightarrow \mathbb{R}$, there exists constants c_i , $i \in [d]$, and strictly increasing continuous functions $\phi_j : [0, 1] \rightarrow [0, 1]$, $j \in [2d + 1]$, and $\Psi \in C[0, 1]$ such that

$$f(x_1, \dots, x_d) = \sum_{j=1}^{2d+1} \Psi \left(\sum_{i=1}^d c_i \phi_j(x_i) \right). \quad (1.6)$$

Rough Idea. We list only a few key steps.

1. Given some $\lambda \in (0, 1)$. Let U_f be the set of open tubes $(\phi_j)_{j \in [2d+1]} \subset C([0, 1])^{\otimes (2d+1)}$ such that there exists some continuous g with uniform bound $\|g\| \leq \mu \|f\|$, $\mu \in (0, 1)$ and

$$\left\| f - \sum_{j=1}^{2d+1} g \left(\sum_{i=1}^d c_i \phi_j(x_i) \right) \right\| \leq \lambda \|f\|.$$

The set U_f is an open, dense subset for any continuous f .

2. Choose a (dense) sequence of continuous functions h_k , and the corresponding sets U_{h_k} . Then, $\bigcap_k U_{h_k}$ is an open, dense subset. One can choose a specific set of tubes $(\phi_j^*)_{j \in [2d+1]} \subset C([0, 1])^{\otimes (2d+1)}$. Then for any continuous

f , there exists h_m arbitrarily close, and corresponding continuous g_m ,

$$\begin{aligned} & \left\| f - \sum_{j=1}^{2d+1} g \left(\sum_{i=1}^d c_i \phi_j^*(x_i) \right) \right\| \\ & \leq \|f - h_m\| + \left\| h_m - \sum_{j=1}^{2d+1} g_m \left(\sum_{i=1}^d c_i \phi_j^*(x_i) \right) \right\| \\ & \leq (\varepsilon + \lambda) \|h_m\| \leq \frac{(\varepsilon + \lambda)}{1 - \varepsilon} \|f\|. \end{aligned}$$

3. Finally, take such a g function iteratively, and construct a sequence that converges to f .

□

Remark 1.2.5. *There are various versions of KST, but the representation size of $2d + 1$ terms cannot be improved.*

1.2.2 Two-Hidden Layers

The MLP model allows the neural network to approximate a continuous function with arbitrary accuracy if the activation function is slightly *pathological*.

Theorem 1.2.6 (Maiorov and Pinkus (1999)). *There exists an activation function $\sigma \in C^\infty$, strictly increasing, sigmoidal, and for any $f \in C[0, 1]^d$ that*

$$\inf_{\theta} \sup_{x \in [0, 1]^d} \left| f(x) - \sum_{i=1}^{4d+3} W_i \sigma \left(\sum_{j=1}^{2d+1} c_{ij} \sigma(w_{ij} \cdot x + b_{ij}) + B_i \right) \right| = 0.$$

Rough Idea. This theorem leverages the pathological activation function to rewrite the KST. Note that the number of inner/outer functions might be improved.

The continuous function Ψ can be approximated by (see Theorem 1.1.45)

$$\Psi(t) \approx a\sigma(t + s_\Psi) + b\sigma(t + s_0) + c\sigma(t + s_1),$$

where σ is the pathological activation function, the shifts s_Ψ depends on Ψ , the other two shifts s_0, s_1 are fixed for linear functions.

Then, replace t with $\sum_i c_i \phi_j(x_i)$ for each $j \in [2d + 1]$, we can obtain $2d + 1$ functions (depends on Ψ), $2d + 1$ linear functions (shift s_0), and $2d + 1$ linear

functions (shift s_1). The $2d + 1$ linear functions can be merged into $d + 1$ linear functions by a linear basis. Thus, in total we need $2d + 1 + 2(d + 1) = 4d + 3$ outer functions.

Similar ideas apply to the inner functions. □

Remark 1.2.7. We can make $\{w_{ij}\}_{i \in [2d+1]}$ explicit. There are d of them as standard base vectors and $d + 1$ of them as unit vectors (for a linear basis)

Remark 1.2.8. Of course, such an approximation cannot be stable, which means finding the parameterization θ could be very difficult.

Extended Reading

Barron, A. (1993). Universal approximation bounds for superpositions of a sigmoidal function. *IEEE Transactions on Information Theory*, 39(3):930–945.

Barron, A. R. (1994). Approximation and estimation bounds for artificial neural networks. *Machine learning*, 14(1):115–133.

Canuto, C. and Quarteroni, A. (1982). Approximation results for orthogonal polynomials in sobolev spaces. *Mathematics of Computation*, 38(157):67–86.

DeVore, R. A., Howard, R., and Micchelli, C. (1989). Optimal nonlinear approximation. *Manuscripta mathematica*, 63(4):469–478.

DLMF (2025). *NIST Digital Library of Mathematical Functions*. F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.

Guo, B.-y., Shen, J., and Xu, C.-l. (2005). Generalized laguerre approximation and its applications to exterior problems. *Journal of computational Mathematics*, pages 113–130.

Hornik, K., Stinchcombe, M., and White, H. (1989). Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366.

Jones, L. K. (1992). A simple lemma on greedy approximation in hilbert space and convergence rates for projection pursuit regression and neural network training. *The annals of Statistics*, pages 608–613.

Maierov, V. and Pinkus, A. (1999). Lower bounds for approximation by mlp neural networks. *Neurocomputing*, 25(1-3):81–91.

Maierov, V. E. (1999). On best approximation by ridge functions. *Journal of Approximation Theory*, 99(1):68–94.

- Makovoz, Y. (1996). Random approximants and neural networks. *Journal of Approximation Theory*, 85(1):98–109.
- Mhaskar, H. N. (1996). Neural networks for optimal approximation of smooth and analytic functions. *Neural computation*, 8(1):164–177.
- Petrushev, P. P. (1998). Approximation by ridge functions and neural networks. *SIAM Journal on Mathematical Analysis*, 30(1):155–189.
- Siegel, J. W. and Xu, J. (2024). Sharp bounds on the approximation rates, metric entropy, and n -widths of shallow neural networks. *Foundations of Computational Mathematics*, 24(2):481–537.

Chapter 2

Training Algorithms

In mathematics, the art of proposing a question must be held of higher value than solving it.

– Georg Cantor



HIS chapter will provide background on the training algorithm for neural networks, including several topics in applications. Since the loss function is generally non-convex, the theoretical studies are more qualitative than quantitative.

2.1 Gradient Descent Method

The training of neural networks relies on optimization techniques. Gradient descent methods iteratively adjust the parameters to minimize the loss function.

Let the loss function be $\mathcal{L}(\theta)$. In the ideal continuum regime, the gradient flow (with infinitesimal learning rate) shows

$$\frac{d}{dt}\theta = -\nabla\mathcal{L}(\theta), \quad \theta(0) = \theta_0.$$

Therefore (if the ODE solves uniquely),

$$\frac{d}{dt}\mathcal{L}(\theta) = -\nabla\mathcal{L}(\theta) \cdot \frac{d}{dt}\theta = -\|\nabla\mathcal{L}(\theta)\|^2 \leq 0.$$

It shows that, as long as $\nabla\mathcal{L} \neq 0$, the gradient flow decreases the loss function. In the discrete setting, the gradient descent method becomes

$$\theta_{k+1} = \theta_k - \eta \nabla\mathcal{L}(\theta_k).$$

The parameter η denotes the learning rate (i.e., the time step size). It need not be constant. For simplicity, let us assume it is for now.

2.1.1 Convexity

Definition 2.1.1. A function f is convex on domain Ω if for any $a \in [0, 1]$,

$$af(x) + (1-a)f(y) \leq f(ax + (1-a)y), \quad \forall x, y \in \Omega.$$

If f is convex and differentiable, then (expand the RHS at $a \approx 0$), we have

$$f(x) \geq \nabla f(y) \cdot (x - y) + f(y).$$

It means the function's graph is above any tangent planes.

2.1.2 L-smoothness

Definition 2.1.2. If $\mathcal{L}$ is differentiable and satisfies

$$\langle \nabla\mathcal{L}(x) - \nabla\mathcal{L}(y), x - y \rangle \leq L\|x - y\|_2^2,$$

then $\nabla\mathcal{L}$ is L -Lipschitz continuous or $\mathcal{L}$ is L -smooth.

Theorem 2.1.3. If $\mathcal{L} \in C^1$ and $\nabla \mathcal{L}$ is L -Lipschitz, then

$$\mathcal{L}(\theta') \leq \mathcal{L}(\theta) + \langle \nabla \mathcal{L}(\theta), (\theta' - \theta) \rangle + \frac{L}{2} \|\theta' - \theta\|^2$$

Question 2.1.4. Prove the above theorem.

The next theorem is a useful argument.

Theorem 2.1.5 (Co-Coercivity). If $\mathcal{L}$ is convex and $\nabla \mathcal{L}$ is L -Lipschitz, then

$$\langle \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y), x - y \rangle \geq \frac{1}{L} \|\nabla \mathcal{L}(x) - \nabla \mathcal{L}(y)\|^2.$$

Proof. Use Theorem 2.1.3,

$$\begin{aligned} \inf_{\theta'} \mathcal{L}(\theta') &\leq \inf_{\theta'} \mathcal{L}(\theta) + \langle \nabla \mathcal{L}(\theta), (\theta' - \theta) \rangle + \frac{L}{2} \|\theta' - \theta\|^2 \\ &= \mathcal{L}(\theta) - \frac{1}{2L} \|\nabla \mathcal{L}(\theta)\|^2. \end{aligned}$$

The above inequality only uses the L -Lipschitz property of the gradient. Let

$$g(\theta) := \mathcal{L}(\theta) - \langle \nabla \mathcal{L}(\theta'), \theta - \theta' \rangle \geq \mathcal{L}(\theta').$$

Now, ∇g is L -Lipschitz, we use the convexity (optimality holds at $\theta = \theta'$), then use the above inequality,

$$\begin{aligned} \mathcal{L}(\theta') &= \inf_{\theta} g(\theta) \leq g(\theta) - \frac{1}{2L} \|\nabla g(\theta)\|^2 \\ &= \mathcal{L}(\theta) - \langle \nabla \mathcal{L}(\theta'), \theta - \theta' \rangle - \frac{1}{2L} \|\nabla \mathcal{L}(\theta) - \nabla \mathcal{L}(\theta')\|^2. \end{aligned}$$

Switch θ and θ' , then combine. □

Problem 2.1.6. Let $\mathcal{L}$ be convex and L -smooth. Suppose θ^* is the global minimizer. Prove that

$$1. \quad \frac{1}{L} \|\nabla \mathcal{L}(\theta_k)\|^2 \leq \langle \theta_k - \theta^*, \nabla \mathcal{L}(\theta_k) \rangle.$$

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2. $\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) + \frac{1}{2L} \|\nabla \mathcal{L}(\theta_k)\|^2 \leq \langle \theta_k - \theta^*, \nabla \mathcal{L}(\theta_k) \rangle.$
3. $\langle \theta_k - \theta^*, \nabla \mathcal{L}(\theta_k) \rangle \leq \mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) + \frac{L}{2} \|\theta_k - \theta^*\|^2$
4. $\mathcal{L}(\theta_k) - \mathcal{L}(\theta_{k-1}) + \frac{1}{2L} \|\nabla \mathcal{L}(\theta_k) - \nabla \mathcal{L}(\theta_{k-1})\|^2 \leq \langle \theta_k - \theta_{k-1}, \nabla \mathcal{L}(\theta_k) \rangle.$
5. $\mathcal{L}(\theta_k) - \mathcal{L}(\theta_{k-1}) \leq \langle \theta_k - \theta_{k-1}, \nabla \mathcal{L}(\theta_k) \rangle.$

Which statements use convexity, and which statements use L -smoothness?

Problem 2.1.7. Find out the continuous version of Problem 2.1.6.

Based on the Theorem 2.1.3, we find that

$$\begin{aligned} \mathcal{L}(\theta_{k+1}) - \mathcal{L}(\theta_k) &\leq \langle \nabla \mathcal{L}(\theta_k), \theta_{k+1} - \theta_k \rangle + \frac{L}{2} \|\theta_{k+1} - \theta_k\|_2^2 \\ &= -\eta \|\nabla \mathcal{L}(\theta_k)\|^2 + \frac{L\eta^2}{2} \|\nabla \mathcal{L}(\theta_k)\|^2 = \left(\frac{L}{2} \eta - 1 \right) \eta \|\nabla \mathcal{L}(\theta_k)\|^2. \end{aligned}$$

Therefore, we need $\eta < \frac{2}{L}$ to make the loss decay. The decay on the right-hand side is maximized when $\eta = \frac{1}{L}$. In general, we only consider $\eta \leq \frac{1}{L}$ for the following reason.

Theorem 2.1.8. Let $\mathcal{L}$ be convex and $\nabla \mathcal{L}$ be L -Lipschitz. If θ^* is the global minimizer, then

$$\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) \leq \frac{1}{2\eta} (\|\theta_k - \theta^*\|^2 - \|\theta_{k+1} - \theta^*\|^2) + \frac{\eta - \frac{1}{L}}{2} \|\nabla \mathcal{L}(\theta_k)\|^2.$$

Proof. Combine

$$\|\theta_{k+1} - \theta^*\|^2 = \|\theta_k - \theta^*\|^2 + \eta^2 \|\nabla \mathcal{L}(\theta_k)\|^2 - 2\eta \langle \nabla \mathcal{L}(\theta_k), \theta_k - \theta^* \rangle$$

with co-coercivity (see last equation of Theorem 2.1.5),

$$\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) + \frac{1}{2L} \|\nabla \mathcal{L}(\theta_k)\|^2 \leq \langle \nabla \mathcal{L}(\theta_k), \theta_k - \theta^* \rangle.$$

Then,

$$\|\theta_{k+1} - \theta^*\|^2 \leq \|\theta_k - \theta^*\|^2 + (\eta^2 - \frac{\eta}{L}) \|\nabla \mathcal{L}(\theta_k)\|^2 - 2\eta (\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)).$$

□

Remark 2.1.9. The existence of a finite global minimizer does not follow from convexity and Lipschitz continuity. For instance, the function $f(x) = -\log x$ is convex on $(1, \infty)$, but the global minimizer does not exist.

Corollary 2.1.10. Let $\mathcal{L}$ be convex and $\nabla \mathcal{L}$ be L -Lipschitz. If θ^* is the global minimizer and $L\eta \leq 1$, then

$$\|\theta_k - \theta^*\| \geq \|\theta_{k+1} - \theta^*\|.$$

We can further determine the convergence rate for $\mathcal{L}(\theta_k)$ (not for θ_k).

Theorem 2.1.11. Let $\mathcal{L}$ be convex and $\nabla \mathcal{L}$ be L -Lipschitz. If θ^* is the global minimizer and $L\eta \leq 1$, then

$$\mathcal{L}(\theta_T) - \mathcal{L}(\theta^*) \leq \frac{1}{2\eta(T+1)} \|\theta_0 - \theta^*\|^2.$$

Proof. The standard method averages $\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)$, that is,

$$\begin{aligned} \mathcal{L}(\theta_T) - \mathcal{L}(\theta^*) &\leq \frac{1}{T+1} \sum_{i=0}^T \mathcal{L}(\theta_i) - \mathcal{L}(\theta^*) \\ &= \frac{1}{T+1} \sum_{i=0}^T \frac{1}{2\eta} (\|\theta_i - \theta^*\|^2 - \|\theta_{i+1} - \theta^*\|^2) \\ &\leq \frac{1}{T+1} \cdot \frac{1}{2\eta} \|\theta_0 - \theta^*\|^2. \end{aligned}$$

□

Remark 2.1.12. Consider the loss function $\mathcal{L}(\theta) = \frac{1}{2p(2p-1)} \theta^{2p}$ on $[-1, 1]$, its gradient is 1-Lipschitz, thus we take $\eta = 1$. The gradient descent becomes

$$\theta_{k+1} = \theta_k - \frac{\theta_k^{2p-1}}{2p-1}.$$

We show that if $\theta_1 = 1$, then

$$\theta_k \asymp k^{-\frac{1}{2p-2}}.$$

The loss function $\mathcal{L}(\theta_k) \asymp k^{-\frac{2p}{2p-2}}$. It shows that $\mathcal{O}(k^{-1})$ is sort of optimal as $p \rightarrow \infty$.

Remark 2.1.13. If $\mathcal{L}$ is just convex and $\nabla \mathcal{L}$ is not continuous, and $\nabla \mathcal{L}$ is just a sub-gradient, the convergence rate becomes $\mathcal{O}(k^{-1/2})$.

Question 2.1.14. Show the gradient flow (suppose the solution is unique) for a convex $\mathcal{L}(\theta)$ satisfying

$$\mathcal{L}(\theta(t)) - \mathcal{L}(\theta^*) \leq \frac{\|\theta_0 - \theta^*\|^2}{2t}.$$

The GD (gradient descent) method is a special case of first-order methods, that is, only gradient information is used:

$$\theta_k \in \theta_0 + \text{span}\{\nabla \mathcal{L}(\theta_0), \dots, \nabla \mathcal{L}(\theta_{k-1})\}.$$

It is known (see [Nesterov et al. \(2018\)](#)) that first-order methods generally can attain at most second-order convergence without further assumptions (other than convexity, L-smoothness, and existence of a finite global minimizer).

Theorem 2.1.15 (Nesterov). For any $1 \leq k \leq \frac{1}{2}(d-1)$ and any $\theta_0 \in \mathbb{R}^d$, there exists a convex loss function $\mathcal{L} : \mathbb{R}^d \rightarrow \mathbb{R}$ with L -Lipschitz gradient $\nabla \mathcal{L}$ that any first-order method leads to

$$\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) \geq \frac{3L\|\theta_0 - \theta^*\|^2}{32},$$

and

$$\|\theta_k - \theta^*\|^2 \geq \frac{1}{8}\|\theta_0 - \theta^*\|^2.$$

A proof can be found in ([Nesterov et al., 2018](#), Chapter 2). The second-order convergence will be introduced in the momentum methods.

2.1.3 Strong Convexity

Definition 2.1.16. $\mathcal{L}$ is μ -strongly convex if $\mathcal{L}(\theta) - \frac{\mu}{2}\|\theta\|^2$ is convex.

The μ -strongly convexity and L -Lipschitz is equivalent to say

$$\mathcal{J}(\theta) := \mathcal{L}(\theta) - \mu \frac{\|\theta\|^2}{2}$$

is $(L - \mu)$ -Lipschitz and convex.

Problem 2.1.17. Show that if $\mathcal{L}$ is μ -strongly convex, then

$$\mathcal{L}(\theta') \geq \mathcal{L}(\theta) + \nabla \mathcal{L}(\theta) \cdot (\theta' - \theta) + \frac{\mu}{2} \|\theta' - \theta\|^2.$$

If θ^* is a global minimizer, then

$$\mathcal{L}(\theta) - \mathcal{L}(\theta^*) \leq \frac{1}{2\mu} \|\nabla \mathcal{L}(\theta)\|^2.$$

Question 2.1.18. Prove that the gradient flow now satisfies

$$\|\theta(t) - \theta^*\|^2 \leq \exp(-\mu t) \|\theta_0 - \theta^*\|^2.$$

Theorem 2.1.19 (Co-Coercivity). The μ -strongly convexity and L -Lipschitz imply the following inequality.

$$\langle \nabla \mathcal{L}(\theta) - \nabla \mathcal{L}(\theta'), \theta - \theta' \rangle \geq \frac{L\mu}{L + \mu} \|\theta - \theta'\|^2 + \frac{1}{L + \mu} \|\nabla \mathcal{L}(\theta) - \nabla \mathcal{L}(\theta')\|^2.$$

Proof. Since $\mathcal{L}(\theta) - \frac{\mu}{2} \|\theta\|^2$ is convex and its gradient is $(L - \mu)$ -Lipschitz, the co-coercivity implies

$$\langle \nabla \mathcal{L}(\theta) - \nabla \mathcal{L}(\theta'), \theta - \theta' \rangle - \mu \|\theta - \theta'\|^2 \geq \frac{\|\nabla \mathcal{L}(\theta) - \nabla \mathcal{L}(\theta') - \mu(\theta - \theta')\|^2}{(L - \mu)}.$$

Rearrange the terms, and obtain

$$(L + \mu) \langle \nabla \mathcal{L}(\theta) - \nabla \mathcal{L}(\theta'), \theta - \theta' \rangle \geq \|\nabla \mathcal{L}(\theta) - \nabla \mathcal{L}(\theta')\|^2 + \mu L \|\theta - \theta'\|^2.$$

□

Theorem 2.1.20. Let $\mathcal{L}$ be μ -strongly convex and $\nabla \mathcal{L}$ be L -Lipschitz. If $\eta \leq \frac{2}{L + \mu}$, then

$$\|\theta_{k+1} - \theta^*\|^2 \leq \left(1 - \eta \frac{2\mu L}{\mu + L}\right) \|\theta_k - \theta^*\|^2.$$

Proof. Use the co-coercivity,

$$\begin{aligned}
 \|\theta_{k+1} - \theta^*\|^2 &= \|\theta_k - \eta \nabla \mathcal{L}(\theta_k) - \theta^*\|^2 \\
 &= \|\theta_k - \theta^*\|^2 - 2\eta \langle \theta_k - \theta^*, \nabla \mathcal{L}(\theta_k) \rangle + \eta^2 \|\nabla \mathcal{L}(\theta_k)\|^2 \\
 &\leq \left(1 - \frac{2\eta L \mu}{L + \mu}\right) \|\theta_k - \theta^*\|^2 + \left(\eta^2 - \frac{2\eta}{L + \mu}\right) \|\nabla \mathcal{L}(\theta_k)\|^2.
 \end{aligned}$$

□

We can see from the above theorem that $\mu > 0$ is essential for exponential decay.

Corollary 2.1.21. *Let $\mathcal{L}$ be μ -strongly convex and $\nabla \mathcal{L}$ be L -Lipschitz. If $\eta \leq \frac{2}{L+\mu}$, then*

$$\|\theta_k - \theta^*\|^2 \leq \left(1 - \eta \frac{2\mu L}{\mu + L}\right)^k \|\theta_0 - \theta^*\|^2.$$

When $\eta = \frac{2}{L+\mu}$, the convergence becomes $\|\theta_k - \theta^*\| \leq \rho^k \|\theta_0 - \theta^*\|$, where

$$\rho = \sqrt{1 - \frac{4\mu L}{(\mu + L)^2}} = \frac{(L - \mu)}{(L + \mu)} = \frac{\kappa - 1}{\kappa + 1}, \quad \kappa = \frac{L}{\mu}.$$

Example 2.1.22. *If the loss function is $\mathcal{L}(\theta) = \frac{1}{2} \theta^T A \theta$, where A is a positive definite matrix. Then $\mu = \min(\lambda(A))$ and $L = \max(\lambda(A))$. The parameter κ represents the condition number. In this case, we can see $\eta = \left(\frac{1}{2}(\mu + L)\right)^{-1}$ is the optimal value for*

$$\min_{\eta \in \mathbb{R}} \max(|1 - \eta\mu|, |1 - \eta L|).$$

2.1.4 Non-convex Assumption

Standard loss functions (e.g., MSE, cross-entropy) for neural networks are generally nonconvex. Therefore, it is not guaranteed that optimization will converge to the global minimum.

Definition 2.1.23. *We consider a sequence of learning rates η_k that satisfies*

$$\sum_{k=1}^{\infty} \eta_k = \infty, \quad \sum_{k=1}^{\infty} \eta_k^2 < \infty.$$

The constant learning rate in the previous section does not belong to this class. The learning rates with slow decay $\eta_k = \frac{C_0}{k+C_1}$ belong to this class. It is proved in Bertsekas and Tsitsiklis (2000) that the gradient sequence $\nabla \mathcal{L}(\theta_k)$ vanishes as $k \rightarrow \infty$ for the gradient descent method with step size η_k generically. The original proof is more general. We provide a much simpler version.

Theorem 2.1.24. *Let $\mathcal{L}(\theta)$ be differentiable and L -smooth, then with the aforementioned learning rates, either $\mathcal{L}(\theta_k) \rightarrow -\infty$, or $\lim_{k \rightarrow \infty} \|\nabla \mathcal{L}(\theta_k)\| = 0$.*

Rough Idea. The L -smoothness implies that

$$\mathcal{L}(\theta_{k+1}) \leq \mathcal{L}(\theta_k) + \nabla \mathcal{L}(\theta_k) \cdot (\theta_{k+1} - \theta_k) + \frac{L^2}{2} \|\theta_{k+1} - \theta_k\|^2.$$

Bring $\theta_{k+1} = \theta_k - \eta_k \nabla \mathcal{L}(\theta_k)$ into the above formula, then

$$\mathcal{L}(\theta_{k+1}) \leq \mathcal{L}(\theta_k) - \left(\eta_k - \frac{L^2}{2} \eta_k^2 \right) \|\nabla \mathcal{L}(\theta_k)\|^2.$$

Choose a small $\varepsilon > 0$. Let us suppose $\eta_k < \varepsilon \ll \frac{2}{L}$ for all $k > k_0$ (the 2nd condition allows us to do so), we can see $\mathcal{L}(\theta_k)$ decays monotonically if $k > k_0$. If

$$\sum_{k > k_0} \eta_k \|\nabla \mathcal{L}(\theta_k)\|^2 = \infty,$$

then $\mathcal{L}(\theta_k) \rightarrow -\infty$. Otherwise, $\lim_{k \rightarrow \infty} \mathcal{L}(\theta_k)$ exists and

$$\sum_{k > k_0} \eta_k \|\nabla \mathcal{L}(\theta_k)\|^2 < \infty.$$

The above bound implies that $\liminf_{k \rightarrow \infty} \|\nabla \mathcal{L}(\theta_k)\| = 0$. Otherwise contradicts with the divergence of $\sum_{k > k_0} \eta_k = \infty$.

It is relatively simple to show $\liminf_{k \rightarrow \infty} \|\nabla \mathcal{L}(\theta_k)\| = 0$ (why?). Next, we show $\limsup_{k \rightarrow \infty} \|\nabla \mathcal{L}(\theta_k)\| = 0$. Otherwise, assume that

$$\limsup_{k \rightarrow \infty} \|\nabla \mathcal{L}(\theta_k)\| = 3\delta > 0.$$

Then, there are infinitely many (sufficiently large) k that $\|\nabla \mathcal{L}(\theta_k)\| \in (\delta, 2\delta)$ for a certain $\delta > 0$ (since the steps are small enough). For any such k , let the

consecutive neighboring indices $\mathcal{N}(k) := \{l \mid \|\nabla \mathcal{L}(\theta_l)\| \in (\delta, 2\delta)\}$. Thus,

$$\sum_{l \in \mathcal{N}(k)} \eta_l \|\nabla \mathcal{L}(\theta_l)\|^2 > \left(\sum_{l \in \mathcal{N}(k)} \eta_l \|\nabla \mathcal{L}(\theta_l)\| \right) \delta \asymp \frac{\delta^2}{L}.$$

where

$$\begin{aligned} \sum_{l \in \mathcal{N}(k)} \eta_l \|\nabla \mathcal{L}(\theta_l)\| &= \sum_{l \in \mathcal{N}(k)} \|\theta_{l+1} - \theta_l\| \\ &\geq \frac{1}{L} \sum_{l \in \mathcal{N}(k)} \left| \|\nabla \mathcal{L}(\theta_{l+1})\| - \|\nabla \mathcal{L}(\theta_l)\| \right| \asymp \frac{\delta}{L}. \end{aligned}$$

Hence, we obtain a contradiction due to the existence of an infinite such k , and $\lim_{k \rightarrow \infty} \|\nabla \mathcal{L}(\theta_k)\| = 0$. $\square$

Remark 2.1.25. It essentially shows that if $f \in \text{Lip}(\mathbb{R})$ that

$$|f(x) - f(y)| \leq L\|x - y\|,$$

and

$$\int_{\mathbb{R}} |f(x)| dx < \infty,$$

Then $\lim_{|x| \rightarrow \infty} f(x) = 0$. Here f can be viewed as $\nabla \mathcal{L}(\theta)$ and $d\theta \sim \theta_{k+1} - \theta_k$.

Question 2.1.26. Does the above theorem 2.1.24 hold with constant learning rates $\eta_k \equiv \eta < \frac{2}{L}$?

Question 2.1.27. What if the learning rates only satisfy a weaker condition $\sum_k \eta_k = \infty$ and $\lim_{k \rightarrow \infty} \eta_k = 0$?

Later, we will see that ℓ^2 -finiteness of the learning rates is useful for stochastic gradient descent.

2.2 Momentum Methods

In this section, we return to the “optimal” convergence rate for first-order methods.

2.2.1 Polyak's method

Polyak's method, or the heavy-ball method, was proposed in Polyak (1964) to overcome the oscillations of the iterates in the usual gradient descent method. The Polyak iteration includes an additional momentum term, which represents a *second-order correction*.

$$\theta_{k+1} = \theta_k - \eta \nabla \mathcal{L}(\theta_k) + \beta(\theta_k - \theta_{k-1}).$$

Note that we need two initial conditions θ_0 and θ_{-1} to compute the iterations. Let θ_k be the position vector of a particle at $t = t_k$, then the above iteration becomes

$$(1 - \beta) \frac{\theta_{k+1} - \theta_k}{\eta} + \beta \eta \left(\frac{(\theta_{k+1} - \theta_k) - (\theta_k - \theta_{k-1})}{\eta^2} \right) = -\nabla \mathcal{L}(\theta_k),$$

It can be *locally* viewed as an approximation to the ODE

$$\frac{d}{dt} \theta + \frac{\beta \eta}{1 - \beta} \frac{d^2}{dt^2} \theta = -\frac{1}{1 - \beta} \nabla \mathcal{L}(\theta).$$

Next, we show that Polyak's method has a better *local* convergence than the standard gradient descent method.

Remark 2.2.1. A high-order ODE can be transformed into a first-order dynamical system. This is a typical treatment. In the above 2nd-order ODE, we can write

$$\frac{d}{dt} \begin{pmatrix} \theta \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & -\frac{1-\beta}{\beta\eta} \end{pmatrix} \begin{pmatrix} \theta \\ \dot{\theta} \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{1}{\beta\eta} \nabla \mathcal{L}(\theta) \end{pmatrix}.$$

Remark 2.2.2. From a physics perspective, the ODE can be viewed as Newton's 2nd law with friction. The variable θ represents the position and $\dot{\theta}$ represents the momentum. The loss function becomes a potential energy. However, it will be hard to explain the standard GD in a similar way using dimension analysis. We will see this in a later section.

Theorem 2.2.3. Set $p_k = \frac{\beta}{1-\beta}(\theta_k - \theta_{k-1})$. Let $\mathcal{L}$ be convex and L -smooth. If θ^* is the global minimizer, then

$$\begin{aligned} \mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) &\leq (1 - \beta) \frac{\|\theta_k + p_k - p^*\|^2 - \|\theta_k + p_{k+1} - p^*\|^2}{2\eta\lambda} \\ &\quad + \frac{(\eta - (1 - \beta)\frac{2-\lambda}{L})}{2(1 - \beta)\lambda} \|\nabla \mathcal{L}(\theta_k)\|^2 \\ &\quad - \frac{\beta}{(1 - \beta)\lambda} (\mathcal{L}(\theta_k) - \mathcal{L}(\theta_{k-1})). \end{aligned}$$

Proof. Due to the similarity of Polyak's method with the standard gradient descent ($\beta = 0$), the proof shares a similar flavor as Theorem 2.1.8. Then

$$\theta_{k+1} + p_{k+1} = \theta_k + p_k - \frac{\eta}{1-\beta} \nabla \mathcal{L}(\theta_k).$$

We have (recall Problem 2.1.6)

$$\begin{aligned} \|\theta_{k+1} + p_{k+1} - \theta^*\|^2 &= \|\theta_k + p_k - \theta^*\|^2 - \frac{2\eta}{1-\beta} \langle \theta_k + p_k - \theta^*, \nabla \mathcal{L}(\theta_k) \rangle \\ &\quad + \left(\frac{\eta}{1-\beta} \right)^2 \|\nabla \mathcal{L}(\theta_k)\|^2 \\ &= \|\theta_k + p_k - \theta^*\|^2 + \left(\frac{\eta}{1-\beta} \right)^2 \|\nabla \mathcal{L}(\theta_k)\|^2 \\ &\quad - \frac{2\eta}{1-\beta} \langle \theta_k - \theta^*, \nabla \mathcal{L}(\theta_k) \rangle \\ &\quad - \frac{2\eta\beta}{(1-\beta)^2} \langle \theta_k - \theta_{k-1}, \nabla \mathcal{L}(\theta_k) \rangle \\ &\leq \|\theta_k + p_k - \theta^*\|^2 + \left(\frac{\eta}{1-\beta} \right)^2 \|\nabla \mathcal{L}(\theta_k)\|^2 \\ &\quad - \frac{2\eta}{1-\beta} \frac{1-\lambda}{L} \|\nabla \mathcal{L}(\theta_k)\|^2 \\ &\quad - \frac{2\eta}{1-\beta} \lambda \left(\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) + \frac{1}{2L} \|\nabla \mathcal{L}(\theta_k)\|^2 \right) \\ &\quad - \frac{2\eta\beta}{(1-\beta)^2} (\mathcal{L}(\theta_k) - \mathcal{L}(\theta_{k-1})) \end{aligned}$$

where $\lambda \in [0, 1]$ is an arbitrary parameter. The conclusion is obtained by rearranging the terms. $\square$

Therefore, if $\eta \leq \frac{2-\lambda}{L}(1-\beta)$, we have

$$\begin{aligned} \mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) &\leq (1-\beta) \frac{\|\theta_k + p_k - \theta^*\|^2 - \|\theta_{k+1} + p_{k+1} - \theta^*\|^2}{2\eta\lambda} \\ &\quad - \frac{\beta}{(1-\beta)\lambda} (\mathcal{L}(\theta_k) - \mathcal{L}(\theta_{k-1})). \end{aligned}$$

Take the sum over k , we find that

$$\begin{aligned} \sum_{k=0}^T \mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) &\leq (1-\beta) \frac{\|\theta_0 + p_0 - \theta^*\|^2}{2\eta\lambda} + \frac{\beta}{(1-\beta)\lambda} (\mathcal{L}(\theta_{-1}) - \mathcal{L}(\theta_T)) \\ &\leq (1-\beta) \frac{\|\theta_0 + p_0 - \theta^*\|^2}{2\eta(1-\beta)\lambda} + \frac{\beta}{(1-\beta)\lambda} (\mathcal{L}(\theta_{-1}) - \mathcal{L}(\theta^*)). \end{aligned}$$

Using the convexity of $\mathcal{L}$, we can replace the LHS by the “moving average”.

Corollary 2.2.4. *If $\eta \leq \frac{2-\lambda}{L}(1-\beta)$, then*

$$\mathcal{L}(\bar{\theta}_T) - \mathcal{L}(\theta^*) \leq \frac{1}{T+1} \left((1-\beta) \frac{\|\theta_0 + p_0 - \theta^*\|^2}{2\eta\lambda} + \frac{\beta(\mathcal{L}(\theta_{-1}) - \mathcal{L}(\theta^*))}{(1-\beta)\lambda} \right),$$

where $\bar{\theta}_T = \frac{1}{T+1} \sum_{k=0}^T \theta_k$ is the Cesáro average.

Remark 2.2.5. *If $\theta_{-1} = \theta_0$ as the initial points, then $p_0 = 0$ and by L -smoothness,*

$$\mathcal{L}(\theta_0) - \mathcal{L}(\theta^*) \leq \frac{L}{2} \|\theta_0 - \theta^*\|^2.$$

The above RHS can be reduced to

$$\frac{1}{2\lambda} \cdot \frac{1}{T+1} \left(\frac{1-\beta}{\eta} + \frac{\beta L}{(1-\beta)} \right) \|\theta_0 - \theta^*\|^2.$$

The optimal β^* satisfies $\beta^* = 1 - \sqrt{\eta L}$.

Theorem 2.2.6. *Take $\lambda = 1$. If $\eta \leq \frac{1}{L}$ and $\beta = 1 - \sqrt{\eta L}$, then*

$$\left[\min_{0 \leq k \leq T} \mathcal{L}(\theta_k) \right] - \mathcal{L}(\theta^*) \leq \frac{1}{2(T+1)} \left(\frac{2\sqrt{\eta L} - \eta L}{\eta} \right) \|\theta_0 - \theta^*\|^2.$$

We can compare the bound for $\beta = 0$, which reduces to $\frac{1}{2(T+1)} \frac{1}{\eta} \|\theta_0 - \theta^*\|^2$ for the standard gradient descent method. The Polyak’s bound is always better since

$$2\sqrt{\eta L} - \eta L - 1 \leq -(1 - \sqrt{\eta L})^2 \leq 0.$$

These two bounds coincide when $\eta = \frac{1}{L}$.

Problem 2.2.7. *Does Polyak’s iteration outperform the standard GD in the sense of the last iteration $\mathcal{L}(\theta_T) - \mathcal{L}(\theta^*)$? What is the main difficulty or difference from the standard GD?*

If we allow variable learning rates and momentum parameters. The convergence can be quantified in the desired way.

Corollary 2.2.8. *Let*

$$\beta_k = \frac{k}{k+1}, \quad \eta_k = \frac{\eta_0}{k+1}.$$

Then we can show that Polyak's method has linear convergence in the standard sense.

$$\mathcal{L}(\theta_T) - \mathcal{L}(\theta^*) \leq \frac{\|\theta_0 - \theta^*\|^2}{2\eta_0(T+1)}.$$

The proof is no different from Theorem 2.2.3 but can be significantly simplified.

Problem 2.2.9. *Prove the above corollary.*

Problem 2.2.10. *Show that the ODE's solution $\theta(t)$ to*

$$(1 - \beta) \frac{d}{dt} \theta + \beta \eta \frac{d^2}{dt^2} \theta = -\nabla \mathcal{L}(\theta)$$

with initial condition $\theta(0) = \theta_0$ and $\theta'(0) = 0$ satisfies

$$\int_0^T (\mathcal{L}(\theta(t)) - \mathcal{L}(\theta^*)) dt \leq \left((1 - \beta) \|\theta_0 - \theta^*\|^2 + \frac{\beta \eta}{1 - \beta} (\mathcal{L}(\theta_0) - \mathcal{L}(\theta^*)) \right).$$

Next, we consider $\mathcal{L}$ as a μ -strongly and L -smooth convex loss function. The following proof is based on [Ghadimi et al. \(2015\)](#). The following simple lemma describes when a sequence decays exponentially.

Lemma 2.2.11 (Exponential Decaying Sequence). *Let $\{A_k\}_{k \geq 0}$ and $\{B_k\}_{k \geq 0}$ be two non-negative sequences. Suppose there are constants $a_1, a_2, b \in \mathbb{R}_+$ and $c \in \mathbb{R}$ that*

$$A_{k+1} + bB_{k+1} \leq a_1 A_k + a_2 A_{k-1} + cB_k,$$

where $A_0 = A_{-1}$, $a_1 + a_2 < 1$, $c < b$. Then

$$A_k \leq \rho^k ((\rho - a_1 + 1)A_0 + bB_0),$$

Continue on next page $\Downarrow$

Continue from last page $\uparrow$

where

$$\rho = \max \left(\frac{c}{b}, \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{2} \right).$$

Proof. Let $r \geq 0$, we construct the relation

$$\begin{aligned} A_{k+1} + rA_k + bB_{k+1} &\leq (a_1 + r)A_k + a_2A_{k-1} + cB_k \\ &\leq \underbrace{(a_1 + r)}_{=\rho} [A_k + rA_{k-1} + bB_k]. \end{aligned}$$

This requires

$$(a_1 + r)r \geq a_2, \quad (a_1 + r)b \geq c.$$

That is,

$$r \geq \max \left(\frac{c}{b} - a_1, \frac{-a_1 + \sqrt{a_1^2 + 4a_2}}{2} \right).$$

Thus,

$$\begin{aligned} A_k &\leq A_k + rA_{k-1} + bB_k \leq \rho^k (A_0 + rA_{-1} + bB_0) \\ &= \rho^k ((1 + \rho - a_1)A_0 + bB_0). \end{aligned}$$

□

The general idea of the proof of convergence in the μ -strongly convex case is similar to that in the previous convex case. The (Lyapunov) functional will be a combination of the loss function and the quadratic distance.

Theorem 2.2.12. Suppose $\mathcal{L}$ is μ -strongly convex and L -smooth. There are constants $\rho \in (0, 1)$ and $C < \infty$ that

$$\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) \leq C\rho^k (\mathcal{L}(\theta_0) - \mathcal{L}(\theta^*)).$$

Proof. We consider the following quantities

$$A_{k+1} = \mathcal{L}(\theta_{k+1}) - \mathcal{L}(\theta^*), \quad B_{k+1} = \|\theta_{k+1} - \theta_k\|^2.$$

We verify that

$$\begin{aligned} \mathcal{L}(\theta_{k+1}) - \mathcal{L}(\theta_k) &\leq \nabla \mathcal{L}(\theta_k) \cdot (\theta_{k+1} - \theta_k) + \frac{L}{2} \|\theta_{k+1} - \theta_k\|^2 \\ &= -\eta \left(1 - \frac{L\eta}{2}\right) \|\nabla \mathcal{L}(\theta_k)\|^2 + \frac{L\beta^2}{2} \|\theta_k - \theta_{k-1}\|^2 \\ &\quad + \beta(1 - \eta L) \nabla \mathcal{L}(\theta_k) \cdot (\theta_k - \theta_{k-1}), \end{aligned}$$

and

$$\|\theta_{k+1} - \theta_k\|^2 = \eta^2 \|\nabla \mathcal{L}(\theta_k)\|^2 + \beta^2 \|\theta_k - \theta_{k-1}\|^2 - 2\eta\beta \nabla \mathcal{L}(\theta_k) \cdot (\theta_k - \theta_{k-1}).$$

The following inequalities are straightforward (from the previous exercises):

$$\begin{aligned} \|\nabla \mathcal{L}(\theta_k)\|^2 &\geq 2\mu(\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)), \\ \nabla \mathcal{L}(\theta_k) \cdot (\theta_k - \theta_{k-1}) &\geq \mathcal{L}(\theta_k) - \mathcal{L}(\theta_{k-1}) + \frac{\mu}{2} \|\theta_k - \theta_{k-1}\|^2. \end{aligned}$$

Suppose we find $b \in \mathbb{R}_+$ that

$$\frac{(1 - \eta L)}{2\eta} < b < \frac{2 - \eta L}{2\eta}.$$

Then (why?)

$$\begin{aligned} &A_{k+1} + bB_{k+1} - A_k - bB_k \\ &= \mathcal{L}(\theta_{k+1}) - \mathcal{L}(\theta_k) + b(\|\theta_{k+1} - \theta_k\|^2 - \|\theta_k - \theta_{k-1}\|^2) \\ &\leq \left(b\eta^2 - \eta\left(1 - \frac{L\eta}{2}\right)\right) \|\nabla \mathcal{L}(\theta_k)\|^2 + \frac{(L + 2b)\beta^2 - 2b}{2} \|\theta_k - \theta_{k-1}\|^2 \\ &\quad + (\beta(1 - \eta L) - 2b\eta\beta) \nabla \mathcal{L}(\theta_k) \cdot (\theta_k - \theta_{k-1}) \\ &\leq 2\mu \left(b\eta^2 - \eta\left(1 - \frac{L\eta}{2}\right)\right) (\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)) + \frac{(L + 2b)\beta^2 - 2b}{2} \|\theta_k - \theta_{k-1}\|^2 \\ &\quad + (\beta(1 - \eta L) - 2b\eta\beta) \left[\mathcal{L}(\theta_k) - \mathcal{L}(\theta_{k-1}) + \frac{\mu}{2} \|\theta_k - \theta_{k-1}\|^2\right] \\ &= (a_1 - 1)A_k + a_2A_{k-1} + (c - b)\|\theta_k - \theta_{k-1}\|^2. \end{aligned}$$

where the constants

$$\begin{aligned} a_1 &= 1 + 2\mu\eta \left(b\eta + \frac{L\eta}{2} - 1\right) + \beta(1 - \eta L - 2b\eta), \\ a_2 &= -\beta(1 - \eta L - 2b\eta), \\ c &= \frac{L + 2b}{2}\beta^2 - \frac{\mu}{2}a_2. \end{aligned}$$

Therefore,

$$A_{k+1} + bB_{k+1} \leq a_1A_k + a_2A_{k-1} + cB_k.$$

and we verify the conditions.

$$a_1, a_2, b > 0, \quad a_1 + a_2 < 1, \quad c < b.$$

The parameter β should satisfy

$$0 \leq \beta < \frac{\mu((L+2b)\eta - 1) + \sqrt{\mu^2((L+2b)\eta - 1)^2 + 4b(L+2b)}}{2(L+2b)},$$

and the ratio

$$\rho = \max \left(\frac{c}{b}, \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{2} \right).$$

Therefore,

$$\mathcal{L}(\theta_{k+1}) - \mathcal{L}(\theta^*) \leq \rho^{k+1}(1 + \rho - a_1)(\mathcal{L}(\theta_0) - \mathcal{L}(\theta^*)).$$

□

Remark 2.2.13. Using the relation that $\mathcal{L}(\theta) - \mathcal{L}(\theta^*) \sim \|\theta - \theta^*\|^2$, this theorem implies the exponential decay of $\|\theta_k - \theta^*\|$. The ODE version of the above theorem can be constructed using a similar idea.

2.2.2 Nesterov's method

Nesterov's method [Nesterov \(1998\)](#) improves on Polyak's method. Instead of the $\mathcal{O}(T^{-1})$ convergence rate for L -smooth convex loss function, Nesterov's method can attain the “optimal” $\mathcal{O}(T^{-2})$ convergence rate using **variable** parameters, which is known as Nesterov Accelerated Gradient (NAG) method. The update scheme is slightly different (and more complicated) from Polyak's momentum method. We first review the case with constant parameters.

$$\begin{aligned} \theta_{k+1} &= \gamma_k - \eta \nabla \mathcal{L}(\gamma_k) \\ \gamma_k &= \theta_k + \underbrace{\beta(\theta_k - \theta_{k-1})}_{\text{momentum}}. \end{aligned} \tag{NAG}$$

Note that there is a difference from Polyak's method

$$\begin{aligned} \theta_{k+1} &= \gamma_k - \eta \nabla \mathcal{L}(\theta_k) \\ \gamma_k &= \theta_k + \underbrace{\beta(\theta_k - \theta_{k-1})}_{\text{momentum}}. \end{aligned} \tag{Heavy-Ball}$$

The iteration γ_k can be viewed as a “lookahead” correction of θ_k in the sense that if γ_k improves the convergence over θ_k , then why not use a more accurate version in the update of gradient descent?

Remark 2.2.14. If we let $v_k = \theta_k - \theta_{k-1}$ be the momentum variable, then the Heavy-

Ball algorithm becomes

$$v_{k+1} = \beta v_k - \eta \nabla \mathcal{L}(\theta_k), \quad \theta_{k+1} = \theta_k + v_{k+1}.$$

The NAG becomes

$$v_{k+1} = \beta v_k - \eta \nabla \mathcal{L}(\underbrace{\theta_k + \beta v_k}_{\text{lookahead point}}), \quad \theta_{k+1} = \theta_k + v_{k+1}.$$

However, the constant parameter setting in NAG still yields linear convergence for convex functions.

Theorem 2.2.15. *Let $\mathcal{L}$ be convex and L -smooth. Assume $\beta \in [0, 1)$, and $\theta_0 = \theta_{-1}$. Then, NAG iteration satisfies*

$$\mathcal{L}(\bar{\gamma}_T) - \mathcal{L}(\theta^*) \leq \frac{1}{T+1} \left(\frac{\beta}{1-\beta} (\mathcal{L}(\gamma_{-1}) - \mathcal{L}(\theta^*)) + \frac{1-\beta}{2\eta} \|\gamma_0 - \theta^*\|^2 \right),$$

where $\bar{\gamma}_T = \frac{1}{T+1} \sum_{k=0}^T \gamma_k$ is the Cesáro average.

Proof. Let $p_k = \frac{\beta}{1-\beta} (\gamma_k - \gamma_{k-1} + \eta \nabla \mathcal{L}(\gamma_{k-1})) = \frac{\beta}{1-\beta} (\gamma_k - \theta_k) = \frac{\beta^2}{1-\beta} (\theta_k - \theta_{k-1})$. Then

$$\begin{aligned} \gamma_{k+1} + p_{k+1} &= \frac{1}{1-\beta} \gamma_{k+1} + \frac{\beta}{1-\beta} (\eta \nabla \mathcal{L}(\gamma_k) - \gamma_k) \\ &= \gamma_k + p_k - \frac{\eta}{1-\beta} \nabla \mathcal{L}(\gamma_k). \end{aligned}$$

This is essentially the same as the proof for Polyak's method. Then, let $\gamma^* = \theta^*$,

recall Problem 2.1.6,

$$\begin{aligned}
\|\gamma_{k+1} + p_{k+1} - \gamma^*\|^2 &= \|\gamma_k + p_k - \gamma^*\|^2 - \frac{2\eta}{1-\beta} \langle \gamma_k + p_k - \gamma^*, \nabla \mathcal{L}(\gamma_k) \rangle \\
&\quad + \frac{\eta^2}{(1-\beta)^2} \|\nabla \mathcal{L}(\gamma_k)\|^2 \\
&= \|\gamma_k + p_k - \gamma^*\|^2 + \frac{\eta^2}{(1-\beta)^2} \|\nabla \mathcal{L}(\gamma_k)\|^2 \\
&\quad - \frac{2\eta}{1-\beta} \langle \gamma_k - \gamma^*, \nabla \mathcal{L}(\gamma_k) \rangle \\
&\quad - \frac{2\beta\eta}{(1-\beta)^2} \langle \gamma_k - \gamma_{k-1}, \nabla \mathcal{L}(\gamma_k) \rangle \\
&\quad - \frac{2\beta\eta^2}{(1-\beta)^2} \langle \nabla \mathcal{L}(\gamma_{k-1}), \nabla \mathcal{L}(\gamma_k) \rangle \\
&\leq \|\gamma_k + p_k - \gamma^*\|^2 + \frac{\eta^2}{(1-\beta)^2} \|\nabla \mathcal{L}(\gamma_k)\|^2 \\
&\quad - \frac{2\eta}{1-\beta} (\mathcal{L}(\gamma_k) - \mathcal{L}(\gamma^*) + \frac{1}{2L} \|\nabla \mathcal{L}(\gamma_k)\|^2) \\
&\quad - \frac{2\beta\eta}{(1-\beta)^2} (\mathcal{L}(\gamma_k) - \mathcal{L}(\gamma_{k-1}) + \frac{1}{2L} \|\nabla \mathcal{L}(\gamma_k) - \nabla \mathcal{L}(\gamma_{k-1})\|^2) \\
&\quad - \frac{2\beta\eta^2}{(1-\beta)^2} \langle \nabla \mathcal{L}(\gamma_{k-1}), \nabla \mathcal{L}(\gamma_k) \rangle
\end{aligned}$$

We rearrange the terms and obtain

$$\begin{aligned}
&\frac{2\eta}{(1-\beta)^2} (\mathcal{L}(\gamma_k) - \mathcal{L}(\gamma^*)) + \|\gamma_{k+1} + p_{k+1} - \gamma^*\|^2 \\
&\leq \frac{2\beta\eta}{(1-\beta)^2} (\mathcal{L}(\gamma_{k-1}) - \mathcal{L}(\gamma^*)) + \|\gamma_k + p_k - \gamma^*\|^2 \\
&\quad + \underbrace{\left(\frac{\eta^2}{(1-\beta)^2} - \frac{\eta}{(1-\beta)L} - \frac{\beta\eta}{(1-\beta)^2} \frac{1}{L} \right)}_{=\frac{\eta(\eta-1/L)}{(1-\beta)^2}} \|\nabla \mathcal{L}(\gamma_k)\|^2 - \frac{\beta\eta}{L(1-\beta)^2} \|\nabla \mathcal{L}(\gamma_{k-1})\|^2 \\
&\quad + \underbrace{\left(\frac{2\beta\eta}{L(1-\beta)^2} - \frac{2\beta\eta^2}{(1-\beta)^2} \right)}_{=\frac{2\beta\eta}{(1-\beta)^2} (\frac{1}{L} - \eta)} \langle \nabla \mathcal{L}(\gamma_{k-1}), \nabla \mathcal{L}(\gamma_k) \rangle
\end{aligned}$$

If $0 < \eta \leq \frac{1}{L}$, the quadratic form of the last 3 terms has a determinant

$$\begin{aligned}\Delta &= \left(\frac{2\beta\eta}{(1-\beta)^2} \left(\frac{1}{L} - \eta \right) \right)^2 + 4 \frac{\eta(\eta - 1/L)}{(1-\beta)^2} \frac{\beta\eta}{L(1-\beta)^2} \\ &= \frac{4\eta(\eta - 1/L)}{(1-\beta)^2} \frac{\beta\eta}{L(1-\beta)^2} (\beta(L\eta - 1) - 1) \leq 0.\end{aligned}$$

Therefore, we take the summation over $k = 0, \dots, T$,

$$\begin{aligned}& \sum_{k=0}^T (\mathcal{L}(\gamma_k) - \mathcal{L}(\gamma^*)) + \sum_{k=0}^T \frac{\beta}{(1-\beta)} (\mathcal{L}(\gamma_k) - \mathcal{L}(\gamma^*)) \\ & \quad + \sum_{k=0}^T \frac{1-\beta}{2\eta} \|\gamma_{k+1} + p_{k+1} - \gamma^*\|^2 \\ & \leq \sum_{k=0}^T \frac{\beta}{(1-\beta)} (\mathcal{L}(\gamma_{k-1}) - \mathcal{L}(\gamma^*)) + \sum_{k=0}^T \frac{1-\beta}{2\eta} \|\gamma_k + p_k - \gamma^*\|^2.\end{aligned}$$

That is,

$$\sum_{k=0}^T (\mathcal{L}(\gamma_k) - \mathcal{L}(\gamma^*)) \leq \frac{\beta}{(1-\beta)} (\mathcal{L}(\gamma_{-1}) - \mathcal{L}(\gamma^*)) + \frac{1-\beta}{2\eta} \|\gamma_0 + p_0 - \gamma^*\|^2.$$

The LHS is bounded below by $(T+1)\mathcal{L}(\bar{\gamma}_T)$ for the Cesáro average. $\square$

Remark 2.2.16. The conclusion is essentially the same as in Polyak's case. The exponential convergence in the μ -strongly convex setting is similar.

Regarding μ -strongly convex functions, the constant parameters $\eta = \frac{1}{L}$ and $\beta = \frac{1-\sqrt{\kappa}}{1+\sqrt{\kappa}}$, $\kappa = \frac{\mu}{L}$ are typical,

$$\begin{aligned}\theta_{k+1} &= \gamma_k - \frac{1}{L} \nabla \mathcal{L}(\gamma_k) \\ \gamma_k &= \theta_k + \frac{1-\sqrt{\kappa}}{1+\sqrt{\kappa}} \underbrace{(\theta_k - \theta_{k-1})}_{\text{momentum}},\end{aligned} \tag{NAG-sc}$$

and the convergence is exponentially fast

$$\mathcal{L}(\theta_T) - \mathcal{L}(\theta^*) \leq \rho^T \left(\mathcal{L}(\theta_0) - \mathcal{L}(\theta^*) + \frac{\mu}{2} \|\theta_0 - \theta^*\|^2 \right), \quad \rho := (1 - \sqrt{\kappa}).$$

This setting is often known as NAG-sc. The proof is based on the construction of a Lyapunov function.

Theorem 2.2.17. *If $\mathcal{L}$ is μ -strongly convex and L -smooth, then NAG-sc converges exponentially,*

$$\mathcal{L}(\theta_T) - \mathcal{L}(\theta^*) \leq \rho^T \left(\mathcal{L}(\theta_0) - \mathcal{L}(\theta^*) + \frac{\mu}{2} \|\theta_0 - \theta^*\|^2 \right).$$

Proof. Let $\mathcal{A}_k = \mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)$, and $\mathcal{B}_k = \|\theta_k - \rho\theta_{k-1} - (1 - \rho)\theta^*\|^2$. Then

$$\begin{aligned} \mathcal{A}_{k+1} &= \mathcal{L}(\theta_{k+1}) - \mathcal{L}(\theta^*) \\ &\leq \mathcal{L}(\gamma_k) - \mathcal{L}(\theta^*) + \langle \nabla \mathcal{L}(\gamma_k), \theta_{k+1} - \gamma_k \rangle + \frac{L}{2} \|\theta_{k+1} - \gamma_k\|^2 \\ &= \mathcal{L}(\gamma_k) - \mathcal{L}(\theta^*) - \frac{L}{2} \|\theta_{k+1} - \gamma_k\|^2 \end{aligned}$$

and use the following to rewrite the first term

$$\begin{aligned} \mathcal{L}(\gamma_k) - \mathcal{L}(\theta_k) &\leq -\langle \nabla \mathcal{L}(\gamma_k), \theta_k - \gamma_k \rangle - \frac{\mu}{2} \|\gamma_k - \theta_k\|^2, \\ \mathcal{L}(\gamma_k) - \mathcal{L}(\theta^*) &\leq -\langle \nabla \mathcal{L}(\gamma_k), \theta^* - \gamma_k \rangle - \frac{\mu}{2} \|\gamma_k - \theta^*\|^2. \end{aligned}$$

We obtain that

$$\begin{aligned} \mathcal{A}_{k+1} + \frac{L}{2} \mathcal{B}_{k+1} &\leq \rho \left(\mathcal{A}_k - \langle \nabla \mathcal{L}(\gamma_k), \theta_k - \gamma_k \rangle - \frac{\mu}{2} \|\gamma_k - \theta_k\|^2 \right) \\ &\quad + (1 - \rho) \left(-\langle \nabla \mathcal{L}(\gamma_k), \theta^* - \gamma_k \rangle - \frac{\mu}{2} \|\gamma_k - \theta^*\|^2 \right) \\ &\quad + \frac{L}{2} \|\theta_{k+1} - \rho\theta_k\|^2 - \frac{L}{2} \|\theta_{k+1} - \gamma_k\|^2 \\ &= \rho(\mathcal{A}_k + \frac{L}{2} \mathcal{B}_k) \\ &\quad + \rho L \langle \theta_{k+1} - \gamma_k, \theta_k - \gamma_k \rangle - \rho \frac{\mu}{2} \|\gamma_k - \theta_k\|^2 \\ &\quad + (1 - \rho) L \langle \theta_{k+1} - \gamma_k, \theta^* - \gamma_k \rangle - (1 - \rho) \frac{\mu}{2} \|\gamma_k - \theta^*\|^2 \\ &\quad + \frac{L}{2} \|\theta_{k+1} - \rho\theta_k - (1 - \rho)\theta^*\|^2 - \frac{L}{2} \|\theta_{k+1} - \gamma_k\|^2 \\ &\quad - \rho \frac{L}{2} \|\theta_k - \rho\theta_{k-1} - (1 - \rho)\theta^*\|^2 \end{aligned}$$

Replace $\theta_k - \rho\theta_{k-1} = (1 - \rho)\theta_k + \frac{\rho}{\beta}(\gamma_k - \theta_k)$, we can compute the remainder part using the prescribed parameters as

$$-\frac{1}{2}(L - \mu) \sqrt{\frac{\mu}{L}} \|\theta_k - \gamma_k\|^2 \leq 0.$$

The verification step can be done symbolically through Mathematica or SymPy.

```
>> rho = 1 - Sqrt[mu/L];
>> beta = (1 - Sqrt[mu/L])/(1 + Sqrt[mu/L]);
>> Simplify[
rho*L*((theta_k1) - (gamma_k))*((theta_k) - (gamma_k))
- rho*(mu/2)*((gamma_k) - (theta_k))^2
+ (1 - rho)*L*((theta_k1) - (gamma_k))*((theta_star) - (gamma_k))
- (1 - rho)*(mu/2)*((gamma_k) - (theta_star))^2
+ (L/2)*((theta_k1) - rho*(theta_k)
- (1 - rho)*(theta_star))^2 - (L/2)*((theta_k1) - (gamma_k))^2
- rho*(L/2)*((1 - rho)*(theta_k) + rho*(1/(beta))*((gamma_k)
- (theta_k)) - (1 - rho)*(theta_star))^2,
Assumptions -> L > 0 && mu > 0]
>> -(1/2) (L-mu) Sqrt[mu/L] (gamma_k-theta_k)^2
```

□

Now, let us consider the NAG-c (Nesterov Accelerated Gradient) method for general **convex** functions with a variable step size.

$$\begin{aligned}\theta_{k+1} &= \gamma_k - \eta \nabla \mathcal{L}(\gamma_k), \\ \gamma_k &= \theta_k + \beta_k(\theta_k - \theta_{k-1}).\end{aligned}\tag{NAG-c}$$

The parameters satisfies $0 < \eta < \frac{1}{L}$ and $\beta_k = \frac{k}{k+3}$. The following theorem implies that NAG-c achieves the optimal convergence rate $\mathcal{O}(T^{-2})$ for convex and L -smooth functions.

Theorem 2.2.18. *Let $\mathcal{L}$ be convex and L -smooth. The NAG-c algorithm satisfies*

$$\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*) \leq \frac{9}{2\eta(k+2)^2} \|\theta_0 - \theta^*\|^2.$$

Proof. For any x , denote $x_+ = x - \eta \nabla \mathcal{L}(x)$, then

$$\begin{aligned}\mathcal{L}(x_+) - \mathcal{L}(x) &\leq \langle \nabla \mathcal{L}(x), x_+ - x \rangle + \frac{L}{2} \|x_+ - x\|^2 \\ &\leq \left(-\frac{1}{\eta} + \frac{L}{2} \right) \|x_+ - x\|^2 \leq -\frac{1}{2\eta} \|x_+ - x\|^2.\end{aligned}$$

Then

$$\begin{aligned}
\mathcal{L}(x^+) - \mathcal{L}(z) &\leq \mathcal{L}(x) - \mathcal{L}(z) - \frac{\eta}{2} \|\nabla \mathcal{L}(x)\|^2 \\
&\leq \langle \nabla \mathcal{L}(x), x - z \rangle - \frac{\eta}{2} \|\nabla \mathcal{L}(x)\|^2 \\
&= -\frac{\eta}{2} \|\nabla \mathcal{L}(x) - \frac{1}{\eta}(x - z)\|^2 + \frac{1}{2\eta} \|x - z\|^2 \\
&= -\frac{1}{2\eta} \|x_+ - z\|^2 + \frac{1}{2\eta} \|x - z\|^2.
\end{aligned} \tag{*}$$

Apply $x = \gamma_k$ and $z = \theta^*$ and $z = \theta_k$ in (*), then by affinity (verify!)

$$\begin{aligned}
&\mathcal{L}(\theta_{k+1}) - ((1 - \beta_k)\mathcal{L}(\theta^*) + \beta_k\mathcal{L}(\theta_k)) \\
&\leq \frac{1}{2\eta} \left(\|\gamma_k - ((1 - \beta_k)\theta^* + \beta_k\theta_k)\|^2 - \|\theta_{k+1} - ((1 - \beta_k)\theta^* + \beta_k\theta_k)\|^2 \right) \\
&= \frac{(1 - \beta_k)^2}{2\eta} \left(\left\| \theta_k + \frac{\beta_k}{1 - \beta_k}(\theta_k - \theta_{k-1}) - \theta^* \right\|^2 - \left\| \theta_{k+1} + \frac{\beta_k}{1 - \beta_k}(\theta_{k+1} - \theta_k) - \theta^* \right\|^2 \right)
\end{aligned}$$

That is,

$$\begin{aligned}
&\frac{\eta}{(1 - \beta_k)^2} (\mathcal{L}(\theta_{k+1}) - \mathcal{L}(\theta^*)) + \frac{1}{2} \left\| \theta_{k+1} + \frac{\beta_k}{1 - \beta_k}(\theta_{k+1} - \theta_k) - \theta^* \right\|^2 \\
&\leq \frac{\beta_k \eta}{(1 - \beta_k)^2} (\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)) + \frac{1}{2} \left\| \theta_k + \frac{\beta_k}{1 - \beta_k}(\theta_k - \theta_{k-1}) - \theta^* \right\|^2 \\
&\leq \frac{\eta}{(1 - \beta_{k-1})^2} (\mathcal{L}(\theta_k) - \mathcal{L}(\theta^*)) + \frac{1}{2} \left\| \theta_k + \frac{\beta_k}{1 - \beta_k}(\theta_k - \theta_{k-1}) - \theta^* \right\|^2
\end{aligned}$$

Since

$$\frac{\beta_k}{(1 - \beta_k)^2} = \frac{k(k+3)}{9} < \frac{(k+2)^2}{9} = \frac{1}{(1 - \beta_{k-1})^2}.$$

Therefore, we have constructed a monotone sequence, and

$$\begin{aligned}
&\frac{\eta}{(1 - \beta_k)^2} (\mathcal{L}(\theta_{k+1}) - \mathcal{L}(\theta^*)) + \frac{1}{2} \left\| \theta_{k+1} + \frac{\beta_k}{1 - \beta_k}(\theta_{k+1} - \theta_k) - \theta^* \right\|^2 \\
&\leq \frac{\eta}{(1 - \beta_0)^2} (\mathcal{L}(\theta_1) - \mathcal{L}(\theta^*)) + \frac{1}{2} \|\theta_0 - \theta^*\|^2 \\
&\leq \frac{1}{2} \|\theta_0 - \theta^*\|^2.
\end{aligned}$$

□

Remark 2.2.19. There are many ways to define suitable sequences β_k as long as

$$\frac{\beta_k}{(1 - \beta_k)^2} \leq \frac{1}{(1 - \beta_{k-1})^2}.$$

Thus, we can choose $\beta_k = \frac{k}{k+2}$ and the convergence constant becomes $\frac{2}{\eta(k+1)^2}$.

The use of NAG-c for strongly convex functions has been extensively studied over the past decade. Early results [Su et al. \(2016\)](#) show that the naive way of ODE formulation of NAG-c cannot converge exponentially fast, while the numerical experiments show exponential convergence. Later, the gap can be closed by adding higher-order correction terms in the ODE formulation.

2.2.3 Lyapunov Functions

In the continuum limit, the previous sections can be viewed from ODE perspectives. Consider the dynamical system

$$\frac{d\theta}{dt} = \phi(\theta(t)).$$

Let $V(\theta(t))$ be a **Lyapunov function** satisfying

$$V(\theta(t)) \geq V^*,$$

where V^* is a lower bound. The time derivative of V along trajectories is

$$\frac{dV}{dt} = \left\langle \nabla V(\theta(t)), \frac{d\theta}{dt} \right\rangle = \langle \nabla V(\theta(t)), \phi(\theta(t)) \rangle \leq 0.$$

Theorem 2.2.20. Assume $V(\theta(t))$ is bounded from below by V^* and

$$-\frac{dV}{dt} \geq W(\theta(t)) \geq 0.$$

Then:

1. $\liminf_{t \rightarrow \infty} W(\theta(t)) = 0$,
2. For any $T > 0$,

$$\min_{0 \leq t \leq T} W(\theta(t)) \leq \frac{1}{T} (V(\theta(0)) - V(\theta(T))) \leq \frac{1}{T} (V(\theta(0)) - V^*).$$

Proof. Since $W(\theta(t)) \leq -\frac{dV}{dt}$, integrating over $[0, T]$ gives

$$\int_0^T W(\theta(t)) dt \leq V(\theta(0)) - V(\theta(T)) \leq V(\theta(0)) - V^*.$$

Hence,

$$T \min_{0 \leq t \leq T} W(\theta(t)) \leq V(\theta(0)) - V^*,$$

which implies the result. Letting $T \rightarrow \infty$ yields $\liminf_{t \rightarrow \infty} W(\theta(t)) = 0$. $\square$

The following theorem is a direct conclusion from convexity.

Theorem 2.2.21. *If $W(\theta)$ is convex, then*

$$W(\bar{\theta}_T) \leq \frac{1}{T} (V(\theta(0)) - V^*),$$

where

$$\bar{\theta}_T := \frac{1}{T} \int_0^T \theta(t) dt.$$

Theorem 2.2.22. *If there exists $\gamma > 0$ such that*

$$W(\theta(t)) \geq \gamma (V(\theta(t)) - V^*),$$

then

$$V(\theta(t)) - V^* \leq (V(\theta(0)) - V^*) e^{-\gamma t}.$$

Proof. We have

$$\frac{dV}{dt} \leq -W(\theta(t)) \leq -\gamma (V(\theta(t)) - V^*).$$

The result follows from Grönwall's inequality. $\square$

Next, we consider the specific example of gradient flow

$$\frac{d\theta}{dt} = -\nabla \mathcal{L}(\theta(t)).$$

Theorem 2.2.23. *If $\mathcal{L} \in C^1$ and $\mathcal{L}(\theta) \geq \mathcal{L}^*$, then*

$$\min_{0 \leq t \leq T} \|\nabla \mathcal{L}(\theta(t))\|^2 \leq \frac{1}{T} (\mathcal{L}(\theta(0)) - \mathcal{L}^*).$$

Proof. Take

$$W(\theta(t)) = \|\nabla \mathcal{L}(\theta(t))\|^2, \quad V(\theta(t)) = \mathcal{L}(\theta(t)).$$

Then

$$\frac{dV}{dt} = \langle \nabla \mathcal{L}(\theta(t)), \dot{\theta}(t) \rangle = -\|\nabla \mathcal{L}(\theta(t))\|^2,$$

Thus take

$$W(\theta(t)) := \|\nabla \mathcal{L}(\theta(t))\|^2.$$

and apply Theorem 2.2.20. $\square$

Theorem 2.2.24. *If*

$$\|\nabla \mathcal{L}(\theta(t))\|^2 \geq 2\mu(\mathcal{L}(\theta(t)) - \mathcal{L}^*),$$

then

$$\mathcal{L}(\theta(t)) - \mathcal{L}^* \leq e^{-2\mu t}(\mathcal{L}(\theta(0)) - \mathcal{L}^*).$$

Proof. Apply Theorem 2.2.22 (exponential decay under $W(\theta) \geq \gamma(V(\theta) - V^*)$) with $\gamma = 2\mu$. $\square$

Theorem 2.2.25. *If $\mathcal{L}$ is convex and $\mathcal{L}^* = \mathcal{L}(\theta^*)$, then*

$$\mathcal{L}(\bar{\theta}_T) - \mathcal{L}^* \leq \frac{1}{T} \|\theta(0) - \theta^*\|^2,$$

where

$$\bar{\theta}_T = \frac{1}{T} \int_0^T \theta(t) dt.$$

Proof. Let $V(\theta) = \|\theta - \theta^*\|^2$. Then

$$\frac{dV}{dt} = 2\langle \theta(t) - \theta^*, \dot{\theta}(t) \rangle = 2\langle \theta(t) - \theta^*, -\nabla \mathcal{L}(\theta(t)) \rangle.$$

By convexity,

$$\mathcal{L}(\theta(t)) - \mathcal{L}^* \leq \langle \nabla \mathcal{L}(\theta(t)), \theta(t) - \theta^* \rangle.$$

Hence

$$\frac{dV}{dt} \leq -2(\mathcal{L}(\theta(t)) - \mathcal{L}^*).$$

Thus

$$-\frac{dV}{dt} \geq W(\theta(t)), \quad W(\theta(t)) := 2(\mathcal{L}(\theta(t)) - \mathcal{L}^*).$$

Applying Theorem 2.2.21 yields

$$\mathcal{L}(\bar{\theta}_T) - \mathcal{L}^* \leq \frac{1}{2T} (V(\theta(0)) - 0) = \frac{1}{2T} \|\theta(0) - \theta^*\|^2.$$

□

Theorem 2.2.26. *If $\mathcal{L}$ is μ -strongly convex, then*

$$\mathcal{L}(\theta(t)) - \mathcal{L}^* \leq e^{-\gamma t} (\mathcal{L}(\theta(0)) - \mathcal{L}^*), \quad \gamma = \frac{\mu}{2}.$$

Proof. Strong convexity implies

$$\mathcal{L}(\theta(t)) - \mathcal{L}^* \geq \frac{\mu}{2} \|\theta(t) - \theta^*\|^2.$$

Using the previous Lyapunov argument with

$$W(\theta(t)) = \mathcal{L}(\theta(t)) - \mathcal{L}^*,$$

and applying Theorem 2.2.22 gives exponential convergence with $\gamma = \mu/2$. □

Next, we consider another example of the second-order system for Polyak's momentum method

$$\ddot{\theta} + a\dot{\theta} + b\nabla\mathcal{L}(\theta) = 0.$$

This can be written as the first-order system

$$\begin{cases} \dot{\theta} = -y, \\ \dot{y} = -ay - b\nabla\mathcal{L}(\theta). \end{cases}$$

Theorem 2.2.27. *Let $\mathcal{L} \in C^2$ and assume $\|\nabla^2\mathcal{L}(\theta)\| \leq L$ (i.e., $\mathcal{L}$ is L -smooth) and $\mathcal{L}(\theta(t)) \geq \mathcal{L}^*$. Suppose $y(0) = 0$, then*

$$\min_{0 \leq t \leq T} \|\nabla\mathcal{L}(\theta(t))\|^2 \leq \frac{1}{T} \frac{a^2 + 2bL}{ab} (\mathcal{L}(\theta(0)) - \mathcal{L}^*).$$

Proof. Let the Lyapunov function be defined as:

$$V(\theta(t)) = \mathcal{L}(\theta(t)) - \mathcal{L}^* + \frac{a}{a^2 + 2bL} \langle \nabla \mathcal{L}(\theta(t)), y \rangle + \frac{L}{a^2 + 2bL} \|y\|^2.$$

Then, using the property of L -smoothness where $\mathcal{L}(\theta) - \mathcal{L}^* \geq \frac{1}{2L} \|\nabla \mathcal{L}(\theta)\|^2$:

$$\begin{aligned} V(\theta(t)) &\geq \frac{1}{2L} \|\nabla \mathcal{L}\|^2 + \frac{a}{a^2 + 2bL} \langle \nabla \mathcal{L}(\theta(t)), y \rangle + \frac{L}{a^2 + 2bL} \|y\|^2 \\ &\geq c_1 (\|\nabla \mathcal{L}(\theta(t))\|^2 + \|y\|^2). \end{aligned}$$

The time derivative is given by:

$$\begin{aligned} -\frac{dV}{dt} &= \langle \nabla \mathcal{L}(\theta(t)), y \rangle - \frac{a}{a^2 + 2bL} \underbrace{\langle \nabla^2 \mathcal{L}(\theta(t)) y, y \rangle}_{\leq L} \\ &\quad - \frac{a}{a^2 + 2bL} \langle \nabla \mathcal{L}(\theta(t)), -ay - b\nabla \mathcal{L} \rangle \\ &\quad - \frac{2L}{a^2 + 2bL} \langle y, -ay - b\nabla \mathcal{L} \rangle. \end{aligned}$$

Bounding the terms:

$$\begin{aligned} -\frac{dV}{dt} &\geq -\langle \nabla \mathcal{L}(\theta(t)), y \rangle - \frac{aL}{a^2 + 2bL} \|y\|^2 + \frac{a^2}{a^2 + 2bL} \langle \nabla \mathcal{L}(\theta(t)), y \rangle \\ &\quad + \frac{ab}{a^2 + 2bL} \|\nabla \mathcal{L}(\theta(t))\|^2 + \frac{2La}{a^2 + 2bL} \|y\|^2 + \frac{2bL \langle y, \nabla \mathcal{L}(\theta(t)) \rangle}{a^2 + 2bL} \\ &= \frac{La}{a^2 + 2bL} \|y\|^2 + \frac{ab}{a^2 + 2bL} \|\nabla \mathcal{L}(\theta(t))\|^2 =: W(\theta(t)). \end{aligned}$$

From this, it follows that:

$$\min_{0 \leq t \leq T} W(\theta(t)) \leq \frac{1}{T} (V(\theta(0)) - V^*)$$

□

Theorem 2.2.28. Suppose $\mathcal{L}$ is convex, then

$$\mathcal{L}(\bar{\theta}(T)) - \mathcal{L}^* \leq \frac{1}{T} \left(\mathcal{L}(\theta(0)) - \mathcal{L}^* + \frac{a^2}{2b} \left\| \theta(0) - \theta^* + \frac{1}{a} y(0) \right\|^2 \right)$$

Proof. Let $V(\theta(t)) = \mathcal{L}(\theta(t)) - \mathcal{L}^* + \frac{a^2}{2b} \|\theta(t) - \theta^* + \frac{1}{a}y\|^2$. And we can check

$$\begin{aligned} \frac{dV(\theta(t))}{dt} &= \langle \nabla \mathcal{L}(\theta(t)), \dot{\theta} \rangle + \frac{a^2}{b} \langle \theta(t) - \theta^* + \frac{1}{a}y, \dot{\theta} + \frac{1}{a}\dot{y} \rangle \\ &= \langle \nabla \mathcal{L}(\theta(t)), y \rangle + \frac{a^2}{b} \langle \theta(t) - \theta^* + \frac{1}{a}y, y + \frac{1}{a}(-ay - b\nabla \mathcal{L}(\theta(t))) \rangle \\ &= \langle \nabla \mathcal{L}(\theta(t)), y \rangle + \frac{a^2}{b} \langle \theta(t) - \theta^* + \frac{1}{a}y, -\frac{b}{a}\nabla \mathcal{L}(\theta(t)) \rangle \\ &= \langle \nabla \mathcal{L}(\theta(t)), y \rangle - a \langle \theta(t) - \theta^*, \nabla \mathcal{L}(\theta(t)) \rangle - \langle y, \nabla \mathcal{L}(\theta(t)) \rangle \\ &= -a \langle \nabla \mathcal{L}(\theta(t)), \theta(t) - \theta^* \rangle \end{aligned}$$

Thus, $W(\theta(t)) = a(\mathcal{L}(\theta(t)) - \mathcal{L}^*)$ and we use Theorem 2.2.21. $\square$

Finally, we consider the Nesterov acceleration case that

$$\ddot{\theta} + \frac{a}{t}\dot{\theta} + b\nabla \mathcal{L}(\theta) = 0$$

Theorem 2.2.29. Suppose $a \geq 3$, then

$$\mathcal{L}(\theta(T)) - \mathcal{L}^* \leq \frac{(a-1)^2}{2b} \frac{\|(\theta(0) - \theta^*)\|^2}{T^2}$$

Proof. Let $V(\theta(t)) = t^2(\mathcal{L}(\theta(t)) - \mathcal{L}^*) + \frac{1}{2b}\|(a-1)(\theta(t) - \theta^*) + ty\|^2$. Then,

$$\begin{aligned} \frac{dV(\theta(t))}{dt} &= 2t(\mathcal{L}(\theta(t)) - \mathcal{L}^*) + t^2 \langle \nabla \mathcal{L}(\theta(t)), y \rangle \\ &\quad + \frac{1}{b} \langle (a-1)(\theta(t) - \theta^*) + ty, -bt\nabla \mathcal{L}(\theta(t)) \rangle \\ &= 2t(\mathcal{L}(\theta(t)) - \mathcal{L}^*) - t \langle (a-1)(\theta(t) - \theta^*), \nabla \mathcal{L}(\theta(t)) \rangle \\ &\leq t(3-a)(\mathcal{L}(\theta(t)) - \mathcal{L}^*). \end{aligned}$$

Therefore, as long as $a \geq 3$, we can choose $W \equiv 0$ and use Theorem 2.2.20,

$$V(\theta(T)) \leq V(\theta(0)).$$

$\square$

Remark 2.2.30. This convergence bound matches the previous NAG-c's constant $\frac{2}{\eta(k+2)^2}$ if we take $a = 3$.

2.2.4 Acceleration vs. Robustness

TODO

2.3 Stochastic Variants

Stochastic variants of training algorithms are essential in practice due to hardware limitations.

Extended Reading

Bertsekas, D. P. and Tsitsiklis, J. N. (2000). Gradient convergence in gradient methods with errors. *SIAM Journal on Optimization*, 10(3):627–642.

Ghadimi, E., Feyzmahdavian, H. R., and Johansson, M. (2015). Global convergence of the heavy-ball method for convex optimization. In *2015 European control conference (ECC)*, pages 310–315. IEEE.

Nesterov, Y. (1998). Introductory lectures on convex programming volume i: Basic course. *Lecture notes*, 3(4):5.

Nesterov, Y. et al. (2018). *Lectures on convex optimization*, volume 137. Springer.

Polyak, B. T. (1964). Some methods of speeding up the convergence of iteration methods. *Ussr computational mathematics and mathematical physics*, 4(5):1–17.

Su, W., Boyd, S., and Candes, E. J. (2016). A differential equation for modeling nesterov’s accelerated gradient method: Theory and insights. *Journal of Machine Learning Research*, 17(153):1–43.

Chapter 3

Convergence Analysis of Training

It's fine to work on any problem, so long as it generates interesting mathematics along the way – even if you don't solve it at the end of the day.

– Andrew Wiles



THIS chapter studies the convergence of training dynamics in neural networks. Several classical regimes are covered.

3.1 Random Feature**3.2 Neural Tangent Kernel****3.3 Mean Field Scaling****3.4 Implicit Bias**

Part II

Foundation

Chapter 4

Reinforcement Learning

Chapter 5

Generative Models